Separable Rank of Radial Kernels in Two and Higher Dimensions: Machine-Checked Proofs in Lean 4 and Isabelle/HOL

GitHub Copilot

Source archive (CC0 1.0): https://github.com/OlliNiemitalo/kernel-rank-lean

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Abstract

A d-variate kernel $K : X_0 \times \cdots \times X_{d-1} \to \mathbb{K}$ over a commutative semiring $\mathbb{K}$ has *separable rank* $\operatorname{srank}_{\mathbb{K}}(K) \in \mathbb{N} \cup \{\infty\}$ equal to the least integer $R \in \mathbb{N}$ for which $K(x) = \sum_{n=0}^{R-1} \prod_{j=0}^{d-1} f_{n,j}(x_j)$ (or ∞ if no finite R exists). On a d-dimensional Cartesian grid of stencil width W, a rank-R kernel evaluates in dRW one-dimensional taps per grid point rather than W^d. While a positive Gaussian $\exp(-\alpha\|x\|^2)$ has separable rank 1, approximating flat-top or oscillatory isotropic kernels—such as defocus disks, Airy point-spread functions, circular low-pass filters, or low-dispersion wave propagators—with finite coordinatewise rank relies on signed profiles built from complex-rate Gaussians and exponential polynomials.

We present a complete machine-checked characterization in Lean 4 (with Mathlib) of the d-way separable rank of continuous circularly and spherically symmetric kernels $K(x) = \varphi(\|x\|^2)$ on $\mathbb{R}^d$ for all $d \geq 2$: (1) $\operatorname{srank}_{\mathbb{R}}(K) < \infty$ if and only if the real profile φ agrees on $[0,\infty)$ with a canonical exponential-polynomial representation $\mathcal{R} = \{(\lambda_u, P_u)\}_{u=1}^m$, $\varphi(s) = \sum_{u=1}^m \operatorname{Re}(P_u(s)e^{\lambda_u s})$, with nonzero polynomials $P_u \in \mathbb{C}[X]$, distinct rates λ_u satisfying $\operatorname{Im}\lambda_u \geq 0$, and $P_u \in \mathbb{R}[X]$ whenever $\operatorname{Im}\lambda_u = 0$; (2) over $\mathbb{C}$, a continuous profile ψ on $[0,\infty)$ satisfies $\operatorname{srank}_{\mathbb{C}}(x \mapsto \psi(\|x\|^2)) = R$ if and only if $\psi(s) = \sum_{\mu\in S} P_\mu(s)e^{\mu s}$ on $[0,\infty)$ for nonzero polynomials $P_\mu \in \mathbb{C}[X]$ at distinct rates $\mu \in S$ with $\sum_{\mu\in S}((d-1)\deg P_\mu + 1) = R$; (3) over $\mathbb{R}$, $\operatorname{srank}_{\mathbb{R}}(x \mapsto \varphi(\|x\|^2)) = C_d(\mathcal{R}) := \sum_{u=1}^m c_d(\lambda_u, P_u)$, where each real-rate component ($\operatorname{Im}\lambda_u = 0$) costs $c_d = (d-1)\deg P_u + 1$ real d-way products (via real Lagrange interpolation) and each nonreal conjugate-pair component ($\operatorname{Im}\lambda_u > 0$) costs $c_d = 2(d-1)\deg P_u + d$ real d-way products (achieved via an alternating root-of-unity polynomial-product identity giving d rather than 2^{d-1} real products for a complex Gaussian when $\deg P_u = 0$, and proved optimal by Alder–Strassen degree peeling over $\mathbb{R}$); and (4) in $d = 2$, scalar extension $\mathbb{R} \hookrightarrow \mathbb{C}$ preserves rank unconditionally and $\operatorname{srank}_{\mathbb{R}}$ equals the degree of the minimal monic half-line differential annihilator $p(D)\varphi_{\mathbb{C}} = 0$, whereas for every $d \geq 3$ each nonreal conjugate pair incurs a degree-independent real-over-complex rank penalty of $d-2$, and every representation $\mathcal{R}$ is the uniform limit on compact sets of multi-Gaussian kernels of real rank at most $B_d(\mathcal{R}) = C_d(\mathcal{R}) - (d-2)\sum_u \deg P_u$ (strictly below exact rank whenever $d \geq 3$ and $\sum_u \deg P_u \geq 1$).

Necessity rests on the continuous one-parameter semigroup theorem $M(t) = \exp(tA)$ on $[0,\infty)$: in a single-theorem case study comparing a 96-line Banach-algebra integral-smoothing proof in Lean 4 against a 256-line finite-difference proof and three finite-dimensional nonstandard proofs (206 and 399 lines in Lean 4, 275 lines in Isabelle/HOL), standard Bochner integration remains shorter and holds in infinite dimensions even after homogeneous scaling $\eta = \varepsilon + \|M(\varepsilon) - 1\|$ removes the 124-line difference-quotient bound, subject to library-snapshot and author-fluency caveats. Finally, we connect the root-of-unity identity to planar harmonic ("beach-ball") differential operators, present illustrative numerical and GPU case studies on an NVIDIA RTX 4070 Ti Super and 14-core CPU (with explicit coefficients in Appendix A, multi-seed/multi-batch variance, and a four-stage conditioning ablation) for 2-D circular lens blur and 3-D homogeneous-room acoustic leapfrog stencils ($R = 1$–9, identifying Kowalczyk and van Walstijn's 27-point IWB scheme as the $R = 1$

real-Gaussian baseline), and document trade-offs against fourth-order compact FDTD and global `cuFFT` methods, wide-stencil boundary obstacles, and open questions. All Lean 4 and Isabelle/HOL formalizations, Python/PyTorch/Triton optimization and GPU benchmark scripts, JSON numerical archives, and figure/manuscript build scripts are publicly available under CC0 1.0 at https://github.com/OlliNiemitalo/kernel-rank-lean.

1 Introduction

Convolving a d-dimensional array with a dense $W \times \cdots \times W$ spatial stencil requires W^d multiply–accumulate operations per output sample. When the kernel $K : \mathbb{R}^d \to \mathbb{K}$ admits a rank-R coordinatewise decomposition

$$K(x_0, \ldots, x_{d-1}) = \sum_{n=0}^{R-1} \prod_{j=0}^{d-1} f_{n,j}(x_j), \tag{1}$$

the d-dimensional convolution reduces to R cascades of d one-dimensional passes, costing dRW operations per sample (or $O(dR)$ operations independent of W when the one-dimensional factors admit recursive causal/anti-causal IIR realizations [19, 78]). Viewing (1) over a commutative semiring $\mathbb{K}$, the *separable rank* $\operatorname{srank}_{\mathbb{K}}(K) \in \mathbb{N} \cup \{\infty\}$ is the least $R \in \mathbb{N}$ for which such a decomposition exists: matrix rank [70] when $d = 2$ and $\mathbb{K}$ is a field, and functional canonical polyadic (CP) or *separation rank* [34, 5, 41, 30] when $d \geq 3$.

For circularly or spherically symmetric (*radial*) kernels in dimension $d \geq 2$,

$$K(x) = \varphi\left(\sum_{j=0}^{d-1} x_j^2\right), \qquad x = (x_0, \ldots, x_{d-1}) \in \mathbb{R}^d,$$

classical characterizations by Maxwell [51] and Kannappan and Sahoo [38] imply that a continuous nonvanishing radial kernel with $\operatorname{srank}_{\mathbb{R}}(K) = 1$ has a Gaussian profile $\varphi(s) = ae^{-\alpha s}$. Real-rate Gaussians ($\alpha > 0$), however, are strictly monotone and convex in the squared radius $s = \|x\|^2$; single Gaussians and nonnegative Gaussian mixtures cannot reproduce flat passbands, sharp cutoffs, or oscillatory diffraction rings [79]. In image processing, real-time graphics, and numerical wave propagation, Niemitalo [56, 59, 60], Garcia [27], and related filter-design studies [46] approximate flat-top and oscillatory radial profiles—such as camera defocus disks, Airy diffraction patterns $(J_1(r)/r)^2$, and ideal circular low-pass filters $\omega_c J_1(\omega_c r)/(2\pi r)$—by sums of *complex-rate* Gaussians $\operatorname{Re}(\beta e^{(-\alpha+i\omega)\|x\|^2})$, whose damped radial oscillations $\cos(\omega\|x\|^2 + \theta)$ synthesize sharp transitions and sidelobe cancellation while maintaining finite separable rank.

1.1 Visual overview: components, rendering terms, and streaming intermediates

Before introducing technical definitions, Figure 1 illustrates in plain visual terms how one-dimensional filters build a circularly symmetric 2-D kernel and why the distinction between *components* and *rendering terms* governs practical GPU execution:

1. **Mathematical "Components" view (Figure 1a):** Replacing a real decay $-\alpha$ by a complex rate $\lambda = -\alpha + i\omega$ preserves the coordinate split $e^{\lambda(x^2+y^2)} = e^{\lambda x^2} e^{\lambda y^2}$ while oscillating radially as $e^{-\alpha r^2}\cos(\omega r^2)$ and $e^{-\alpha r^2}\sin(\omega r^2)$ ($r^2 = x^2 + y^2$). Writing $g(x) = \sqrt{\beta}\, e^{\lambda x^2} = g_{\mathrm{re}}(x) + i g_{\mathrm{im}}(x)$, taking the real part of $g(x)g(y)$ uses two real 1-D×1-D outer products:

$$\operatorname{Re}\big(g(x)g(y)\big) = g_{\mathrm{re}}(x)\, g_{\mathrm{re}}(y) + g_{\mathrm{im}}(x)\, \big(-g_{\mathrm{im}}(y)\big).$$

   Neither outer product is circular on its own—each is a rectangular plaid of positive (blue) and negative (red) stripes—yet adding the pair cancels the grid corners to form a circularly

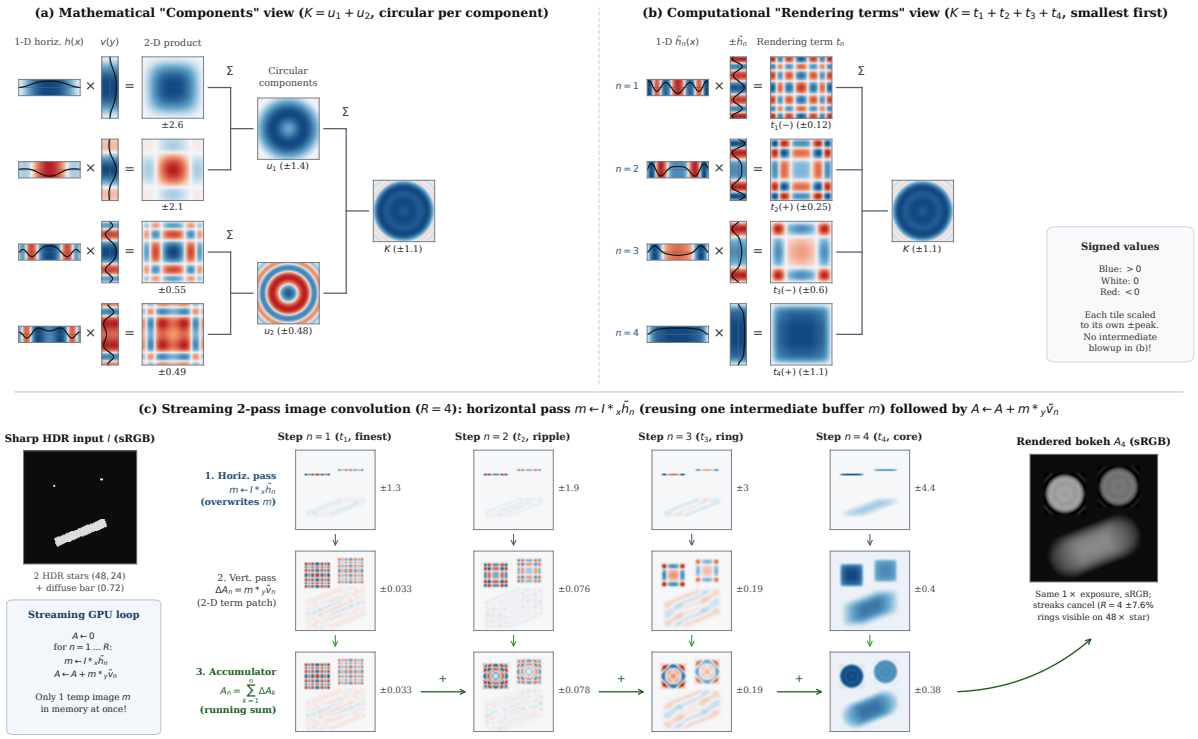


Figure 1: Visual guide to separable radial filtering on a two-complex-Gaussian ($n_{cx} = 2$, exact rank $R = 4$) circular disc-blur kernel [60]: (a) mathematical *Components* view, where pairs of 1-D outer products $h(x)v(y)$ form circularly symmetric concentric-ring components u_1, u_2 that sum to the flat-top disc K; (b) computational *Rendering terms* view, where orthogonalizing rowSpan(K) yields four direct real products $t_n(x, y) = \tilde{h}_n(x) \cdot (\pm\tilde{h}_n(y))$ ordered smallest-first so intermediate amplitudes stay bounded; (c) streaming two-pass linear-light HDR blur using *one* overwritten horizontal intermediate buffer $m \leftarrow I *_x \tilde{h}_n$ followed by vertical accumulation $A \leftarrow A + m *_y \tilde{v}_n$ (with I and A_4 displayed at identical 1× linear exposure through the IEC 61966-2-1 sRGB transfer function; all axis-aligned streaks cancel, while $R = 4$'s ±7.6% pass/stopband ripple remains visible on the 48× HDR star).

symmetric component (u_1: wide dome; u_2: concentric sharpening rings), whose sum $u_1 + u_2$ approximates a flat-top disc K. (When a component multiplies $e^{\lambda r^2}$ by a polynomial $P(r^2)$ of degree k, expanding $(x^2 + y^2)^q$ naively across monomials would produce $(k + 1)(k + 2)$ real products, whereas collecting powers of x^2 merges them into $2(k + 1)$ real products, or $k + 1$ when $\omega = 0$.)

2. **Computational "Rendering terms" view (Figure 1b):** A renderer never computes the circular components u_1, u_2 separately—only the total sum $K(x, y) = \sum_{n=1}^{R} \tilde{h}_n(x)\tilde{v}_n(y)$. In high-accuracy approximations (such as $n_{cx} = 6$ complex Gaussians, $R = 12$), individual components u_i grow hundreds of times taller than K and cancel against each other, causing severe precision loss in 16-bit half-float (`float16`) GPU buffers. Orthogonalizing the 1-D filter subspace (`separableChannels` in [60]; Section 6.3) rotates the R filter pairs into orthogonal *rendering terms* $t_n(x, y) = \tilde{h}_n(x) \cdot (\pm\tilde{h}_n(y))$ ordered from smallest to largest magnitude ($n = 1, \dots, 4$): every term stays comparable to the kernel's own peak (±0.12 to ±1.1 in Figure 1b, vs. ±2.6 in Figure 1a).

3. **Streaming two-pass convolution with one intermediate buffer (Figure 1c):** In the minimal-memory streaming schedule (used, e.g., in fragment-shader pipelines [60]), the GPU initializes an output accumulator $A \leftarrow 0$ and loops over $n = 1, \dots, R$ (smallest term

Table 1: Overview of main theoretical, formalization, and numerical results ($d \geq 2$, degree $k = \deg P$).

Regime	Setting	Main result	Ref.
Finite-rank form	Cont. φ on $\mathbb{R}_{\geq 0}$	$\operatorname{srank}_{\mathbb{R}} < \infty \iff \varphi(s) = \sum_u \operatorname{Re}(P_u(s)e^{\lambda_u s})$ $(s \geq 0)$	Thm. 2.3
Complex rank	$\psi : \mathbb{R} \to \mathbb{C}$, $\mathbb{R}^d$	$(d-1)k+1$ per distinct rate $\mu \in S$	Thm. 2.4
Real rank (Im λ=0)	$P \in \mathbb{R}[X]$, $\mathbb{R}^d$	$c_d(\lambda, P) = (d-1)k+1$ $(= 1$ at $k=0)$	Thm. 2.5
Real rank (Im λ>0)	$P \in \mathbb{C}[X]$, $\mathbb{R}^d$	$c_d(\lambda, P) = 2(d-1)k+d$ $(= d$ vs. 2^{d-1} at $k=0)$	Thm. 2.5, Prop. 2.10
Real–complex gap	Conjugate pair	$c_d(\lambda, P) - 2((d-1)k+1) = d-2$ (all $k \geq 0$)	Rem. 2.13
Border-rank bound	Any list $\mathcal{R}$	Border rank $\leq B_d(\mathcal{R}) = C_d(\mathcal{R}) - (d-2)\sum_u \deg P_u$	Thm. 2.8
2-D diff. order	$d=2$, D on $\mathbb{R}_{\geq 0}$	$\operatorname{srank}_{\mathbb{R}} = \deg p_{\min}$ for minimal $p_{\min}(D)\varphi_{\mathbb{C}} = 0$	Thm. 2.6
Semigroup case study	$M(t) = \exp(tA)$	Integral (96 loc, Banach) vs. 4 FD/NSA (206–399 loc)	Secs. 4–5
Case studies	Blur; 3-D wave	`fp16` sRGB ablation; $\lvert\delta c/c_0\rvert \leq 0.049\%$, vs. FDTD/`cuFFT`	Secs. 6–7, App. A

first). In step n, a 1-D horizontal pass writes $m(x,y) \leftarrow (I *_x \tilde{h}_n)(x,y)$, smearing bright point stars horizontally into 1-D signed streaks; *the same temporary buffer m is overwritten on every step*, keeping DRAM scratch constant at one image regardless of R (in Section 6, `gpu_kernels.py` further tiles the image into L2-cache-resident strips so all R intermediates stay on chip). A 1-D vertical pass then adds $\Delta A_n = m *_y \tilde{v}_n$ into A: steps $n = 1, 2, 3$ build a hollow sharpening ring A_3, and step $n = 4$ adds the soft core so all horizontal and vertical streaks cancel in A_4.

1.2 Plain-language summary of main results and paper organization

Table 1 summarizes our results, organized into three parts so readers can navigate Mathematical Theory (Sections 2–3), Formal Verification (Sections 4–5), and Applications & Limitations (Sections 6–7, Appendix A) independently:

1. **Part I — Mathematical characterization and exact ranks (Sections 2–3):** First, exponential-polynomial profiles in $s = \|x\|^2$ are *universal*: no other continuous radial profile on $\mathbb{R}^d$ $(d \geq 2)$ admits finite coordinatewise separable rank (Theorem 2.3). Second, we establish closed-form exact d-way separable ranks over $\mathbb{C}$ (Theorem 2.4) and over $\mathbb{R}$ (Theorem 2.5), accompanied by a plain-language guide (Section 2.3) and worked $d = 3$ examples for both $k = 0$ and $k = 1$: in $d \geq 3$, an alternating root-of-unity identity reduces each complex Gaussian from 2^{d-1} naive real products to d (3 instead of 4 in 3-D) and each degree-k nonreal component to $2(d-1)k+d$ real products (7 instead of 9 at $d = 3, k = 1$), incurring a constant $d-2$ real-over-complex penalty independent of k. Third, we prove a general multi-Gaussian approximation theorem (Theorem 2.8) showing that every radial profile with $\deg P_u \geq 1$ in $d \geq 3$ has compact-open real border rank strictly smaller than its exact real rank by at least $(d-2)\sum_u \deg P_u$.
2. **Part II — Semigroup case study and formalization architecture (Sections 4–5):** All mathematical theorems in Sections 2–4 (22 top-level Lean 4 theorems and 1 Isabelle/HOL theorem) are machine-checked without unproved steps. Section 4 examines five formalizations of the continuous one-parameter semigroup theorem $M(t) = \exp(tA)$ as an observational case study, introducing a projective normalization $\eta = \varepsilon + \|M(\varepsilon) - 1\|$ that removes the 124-line difference-quotient bound from Euler's infinitesimal proof while discussing why Mathlib's Bochner-integral API keeps standard integration shorter and infinite-dimensional in our development. Section 5 documents the module architecture, reusable vs. problem-specific components, `noncomputable` infimum semantics, and classical axioms.
3. **Part III — Illustrative case studies, limitations, and reproducible tables (Sections 6–7, Appendix A):** Section 6 unifies odd and even planar "beach-ball" differential

operators, proves two-stage steerable pass-count lower bounds, and presents illustrative 2-D lens-blur and 3-D acoustic wave-propagation case studies (including multi-seed `float16` conditioning ablations, multi-batch GPU and CPU timing statistics, and fourth-order compact FDTD and `cuFFT` baselines). Section 7 examines continuous-domain scope, hardware sensitivity, root-of-unity conditioning, and open border-rank questions, while Appendix A tabulates explicit coefficients for standalone reproduction.

1.3 Related work

Classical radial characterizations, continuous Kronecker theorems, and functional equations. Maxwell [51] and Kannappan and Sahoo [38] proved that measurable or continuous rotation-invariant rank-1 functions are Gaussians, while Schoenberg [65] characterized completely monotone radial profiles as positive Gaussian mixtures. Under the coordinate surjection $s_j = x_j^2$, radial separability on $\mathbb{R}^d$ becomes the d-variable additive Hankel functional equation $\varphi(s_0 + \cdots + s_{d-1}) = \sum_{n=0}^{R-1} \prod_{j=0}^{d-1} g_{n,j}(s_j)$ on $[0,\infty)^d$. In two variables, functional equations $f(x+y) = \sum_{n=1}^{R} X_n(x) Y_n(y)$ go back to Levi-Civita [48], Stone [66], Anselone and Korevaar's characterization of finite-dimensional translation-invariant subspaces [3], Laird [45], and Laczkovich [44]. In operator and moment theory, Rochberg [64], Andersson and Carlsson [2], and Mourrain [53] established continuous and multivariate analogues of Kronecker's 1881 theorem connecting finite-rank Hankel and convolution operators to polynomial-exponential symbols and Artinian Gorenstein algebras; whereas bounded L^2 integral-operator formulations assume square-integrability or bounded domains, our pointwise algebraic formulation on $[0,\infty)^d$ characterizes all continuous profiles on $[0,\infty)$—including undamped or polynomially growing profiles such as $\|x\|^2 \cos(\|x\|^2)$—via operator-norm continuous shift semigroups without assuming differentiability or decay.

Separation rank, RBF kernels, and exponential-sum approximations. Beylkin and Mohlenkamp [5], Hackbusch [30], and Khoromskij [39] develop tensor-product and separation-rank representations for multivariate radial operators, building on nonlinear exponential-sum approximation theory [12, 29, 13, 6, 49] and skeleton/cross approximations of smooth radial kernels [71, 75, 69]. Two distinctions clarify how our results complement this literature: (i) a pairwise RBF kernel $(x,y) \in \mathbb{R}^d \times \mathbb{R}^d \mapsto e^{-\alpha\|x-y\|^2}$ has *infinite* exact two-argument rank on $\mathbb{R}^d \times \mathbb{R}^d$ (since $e^{2\alpha x\cdot y}$ is not a finite sum of products $u_n(x)v_n(y)$), whereas as a *convolution stencil* $z \mapsto \varphi(\|z\|^2)$ in $z = x - y$ its coordinatewise d-way rank across $(z_0,\dots,z_{d-1})$ is 1; and (ii) prior sum-of-Gaussians methods construct upper-bound approximations, whereas Theorems 2.3–2.5 establish the exact converse and exact d-way real CP rank for all signed/oscillatory exponential-polynomial profiles.

Algebra multiplicative complexity, Hankel/Waring tensors, real vs. complex gaps, and border rank. Once necessity identifies the $\mathbb{K}[D]$-module $\mathcal{W}(f)$ with the commutative quotient algebra $\mathcal{A} = \mathbb{K}[X]/(p_{\min,\mathbb{K}}(X))$ (where $p_{\min,\mathbb{K}}$ has m_{irr} distinct irreducible factors over $\mathbb{K}$), the multilinear core of our d-way rank problem is the $(d-1)$-fold multiplication tensor $\mathcal{A}^{d-1} \times \mathcal{A}^* \to \mathbb{K}$, $(u_1,\dots,u_{d-1},\ell) \mapsto \ell(u_1 \cdots u_{d-1})$. At $d = 3$ (2-fold bilinear multiplication), Winograd [76], Fiduccia and Zalcstein [23], and Alder and Strassen [1] (with nilradical extensions by Bläser [9]; see Bürgisser, Clausen, and Shokrollahi [15], Thm. 17.20) proved that multiplying in $\mathbb{K}[X]/(p)$ requires $2\dim_{\mathbb{K}} \mathcal{A} - m_{\mathrm{irr}}$ bilinear multiplications, yielding our $d = 3$ costs $2k+1$ and $4k+3$. Recently, Bläser, Mayer, and Shringi [10] generalized the Alder–Strassen lower bound to $(d-1)$-fold multiplication in arbitrary associative algebras, establishing the lower bound $(d-1)\dim_{\mathbb{K}} \mathcal{A} - (d-2)m_{\mathrm{irr}}$; for singly generated commutative algebras over sufficiently large fields, each primary factor attains $(d-1)\dim_{\mathbb{K}} \mathcal{A}_u - (d-2)$ via field-node Lagrange interpolation [10, Cor. 11], summing by Chinese remainder decomposition across the m_{irr} primary

factors. Theorems 2.4 and 2.5 connect continuous radial kernels on $\mathbb{R}^d$ to this invariant, replace ill-conditioned real Vandermonde inversion on nonreal factors $((X+\alpha)^2+\omega^2)^{k+1}$ by closed-form unit-circle root-of-unity functional syntheses (Propositions 2.9–2.10), and machine-check both the lower and upper bounds in Lean 4. This complements apolarity and discrete Hankel/Waring tensor theory [17, 36, 43, 55], real-versus-complex rank separations [18, 20, 63, 11], border-rank gaps on W-type and local-algebra tensors [8, 47, 14, 80], and border-subrank results on higher-order algebra structure tensors [16].

Steerable filters, acoustic FDTD stencils, semigroups, and formalized mathematics. In vision and graphics, multi-stage separable planar filters [70], steerable and complex-derivative filters [26, 40, 61, 7, 62, 73], anisotropic Gaussians [46], and complex-Gaussian circular bokeh filters [56, 27, 57, 58, 59, 60] motivate our analytical and GPU blur studies; in computational acoustics and geophysics, dispersion-relation-preserving and k-space methods [68, 67], compact 27-point and higher-order 3-D room-acoustics FDTD stencils [42, 74, 31, 32], and low-rank wave extrapolators [25] provide our 3-D wave-propagation baselines. Finally, our semigroup and rank formalizations build on classical Banach-algebra semigroups [54, 22], nonstandard Lie theory [33, 28], Lean 4/Mathlib [52, 50, 21], Isabelle/HOL analysis and NSA [24, 35, 72], and formalized tensor-rank developments [4, 37, 77]: whereas Bentkamp, Blanchette, and Klakow [4] formalize CP-rank and generalized matricization bounds for deep convolutional arithmetic circuits in Isabelle/HOL, Jain [37] formalizes lower bounds (≥ 18, via a two-slice real pencil bound) and the Krawczyk principle in Lean 4 (paired with external rational inequality checks for rank ≤ 25) on $8\times 8\times 8$ octonion multiplication, and Yang et al. [77] prove $R_{\mathbb{F}_2}\geq 21$ for 3×3 matrix multiplication over $\mathbb{F}_2$ via finite-geometry certificates in Lean 4, our development treats *infinite-dimensional function spaces* on $\mathbb{R}^d$ and proves parametric closed-form ranks for all dimensions $d\geq 2$ and polynomial degrees $k\geq 0$ (Section 5).

2 Problem Formulation, Harmonized Notation, and Main Theorems

2.1 Separable rank, coordinate surjections, and canonical representations

Throughout the paper, we adhere to one uniform profile convention: $\varphi:\mathbb{R}\to\mathbb{R}$ denotes a *real-valued* radial profile, $\varphi_{\mathbb{C}}(s)=(\varphi(s):\mathbb{C})$ denotes its complexification, $\psi:\mathbb{R}\to\mathbb{C}$ denotes a general *complex-valued* radial profile, and $f:\mathbb{R}\to\mathbb{K}$ denotes a profile taking values in an arbitrary commutative semiring or field $\mathbb{K}$. Similarly, for any kernel $K:X\to\mathbb{R}$, we write $K_{\mathbb{C}}(x)=(K(x):\mathbb{C})$ for its scalar extension to $\mathbb{C}$.

Definition 2.1 (Coordinatewise separable rank; Lean: `IsSumOfProducts`, `separableRank`)**.** Let ι be a finite index set, $(X_i)_{i\in\iota}$ a family of sets, and $\mathbb{K}$ a commutative semiring. A kernel $K:(\prod_{i\in\iota}X_i)\to\mathbb{K}$ is a *sum of R products* ($R\in\mathbb{N}$) if there exist univariate factors $f_{n,i}:X_i\to\mathbb{K}$ ($0\leq n<R$, $i\in\iota$) such that $K(x)=\sum_{n=0}^{R-1}\prod_{i\in\iota}f_{n,i}(x_i)$ for all $x\in\prod_{i\in\iota}X_i$ (taking the empty sum 0 when $R=0$). Its *separable rank* over $\mathbb{K}$ is the extended natural number

$$\operatorname{srank}_{\mathbb{K}}(K):=\inf\bigl\{R\in\mathbb{N}:K\text{ is a sum of }R\text{ products over }\mathbb{K}\bigr\}\in\mathbb{N}\cup\{\infty\},$$

where $\inf\emptyset=\infty$, and an equation $\operatorname{srank}_{\mathbb{K}}(K)=R$ with $R\in\mathbb{N}$ denotes equality in $\mathbb{N}\cup\{\infty\}$ under the canonical inclusion $\mathbb{N}\hookrightarrow\mathbb{N}\cup\{\infty\}$.

Separable rank is subadditive under finite sums and cannot increase when we substitute coordinate maps $y_i\mapsto g_i(y_i)$ into an existing decomposition $\sum_n\prod_i f_{n,i}(g_i(y_i))$. When every $g_i:Y_i\to X_i$ ($i\in\iota$) is *onto* (surjective), the reverse inequality holds as well: picking for each $x_i\in X_i$ at least one preimage $h_i(x_i)\in Y_i$ with $g_i(h_i(x_i))=x_i$ turns any decomposition

Table 2: Harmonized symbol glossary and cost specializations ($d \geq 2$, $k = \deg P$).

Mathematical symbol	Lean 4 identifier	Meaning / specialization
$\operatorname{srank}_{\mathbb{K}}(K) \in \mathbb{N} \cup \{\infty\}$	`separableRank K (ENat)`	Coordinatewise separable rank over $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$
φ, $\varphi_{\mathbb{C}}$, ψ, f	φ : $\mathbb{R} \to \mathbb{R}$, ψ : $\mathbb{R} \to \mathbb{C}$	Real profile, complexification, complex profile, $\mathbb{K}$-profile
$W = 2L + 1$; $s = \|x\|^2$	—; $\sum_j$ `x j ^ 2`	1-D discrete tap width on $[-L, L]$; squared radius on $\mathbb{R}^d$
$\mathcal{W}(f)$	`translateSpan f`	Half-line translate span $\operatorname{span}_{\mathbb{K}}\{u \mapsto f(u+t) : t \geq 0\}$
(λ_u, P_u); $\lambda_u = -\alpha_u + i\omega_u$	`Component.mk rate poly`	Well-formed: $P_u \neq 0$, $\omega_u \geq 0$, $\omega_u{=}0 \Rightarrow P_u \in \mathbb{R}[X]$
n_{re}, n_{cx}; m_{irr}	—	Real and nonreal component counts; $m_{\mathrm{irr}} = n_{\mathrm{re}} + n_{\mathrm{cx}}$
$C_d(\mathcal{R}) = \sum_u c_d(\lambda_u, P_u)$	`multiTotalCost d cs`	Exact d-D real radial rank (Theorem 2.5)
$C_2(\mathcal{R})$; $n_{\mathrm{re}} + d\, n_{\mathrm{cx}}$	`totalCost`; `gaussianTotalCost`	Specializations at $d = 2$ and at $\deg P_u = 0$
$B_d(\mathcal{R}) = \sum_u b_d(\lambda_u, P_u)$	`borderTotalCost d cs`	Multi-Gaussian border bound $C_d(\mathcal{R}) - (d-2)\sum_u \deg P_u$
$H_P(s_0; X)$; $E_k(s_j; X)$	`revHassePoly`; `truncExpPoly`	Reversed Hasse poly.; truncated exp $\sum_{q=0}^{k} s_j^q X^q/q!$
$M^{\mathbb{C}}_{d,k}$, $M^{\mathbb{R}}_{d,k}$	$(d-1)k+1$, $2(d-1)k+d$	Complex and nonreal root-of-unity node counts

$\sum_n \prod_i \tilde{f}_{n,i}(y_i)$ of $K(g_0(y_0), \dots)$ back into $\sum_n \prod_i \tilde{f}_{n,i}(h_i(x_i)) = K(x)$ with the same number of terms:

$$\operatorname{srank}_{\mathbb{K}}\big(y \mapsto K(i \mapsto g_i(y_i))\big) = \operatorname{srank}_{\mathbb{K}}(K). \tag{2}$$

For $x_j \in \mathbb{R}$, the squaring map $x_j \mapsto x_j^2$ covers $\mathbb{R}_{\geq 0} = [0, \infty)$ (with right inverse $s_j \mapsto \sqrt{s_j}$). Applying (2) therefore equates the d-way separable rank of any radial kernel on $\mathbb{R}^d$ with that of the d-way additive Hankel kernel on $\mathbb{R}^d_{\geq 0}$:

$$\operatorname{srank}_{\mathbb{K}}\left(x \in \mathbb{R}^d \mapsto f\left(\sum_{j=0}^{d-1} x_j^2\right)\right) = \operatorname{srank}_{\mathbb{K}}\left(s \in \mathbb{R}^d_{\geq 0} \mapsto f\left(\sum_{j=0}^{d-1} s_j\right)\right). \tag{3}$$

Notice two immediate consequences of (3): (i) because $\|x\|^2 \geq 0$ never visits negative arguments, $K(x) = \varphi(\|x\|^2)$ depends only on $\varphi|_{[0,\infty)}$ and leaves $\varphi(s)$ for $s < 0$ completely unconstrained; and (ii) replacing $x_j \mapsto x_j^2$ by $x_j \mapsto x_j^2/a_j^2$ ($a_j \neq 0$) or $x_j \mapsto |x_j|^p$ ($p > 0$) is still onto $[0, \infty)$, so every theorem below holds verbatim for axis-aligned ellipsoidal kernels $f(\sum_j x_j^2/a_j^2)$ and ℓ_p superellipsoidal kernels $f(\sum_j |x_j|^p)$. Finally, for $d \geq 2$, setting $x_2 = \cdots = x_{d-1} = 0$ turns each d-way product $\prod_{j=0}^{d-1} f_{n,j}(x_j)$ into the 2-way product $(c_n f_{n,0}(x_0)) f_{n,1}(x_1)$ with constant $c_n = \prod_{j=2}^{d-1} f_{n,j}(0)$, giving the 2-D slice lower bound:

$$\operatorname{srank}_{\mathbb{K}}\big(y \in \mathbb{R}^2 \mapsto f(y_0^2 + y_1^2)\big) \leq \operatorname{srank}_{\mathbb{K}}\left(x \in \mathbb{R}^d \mapsto f\left(\sum_{j=0}^{d-1} x_j^2\right)\right). \tag{4}$$

Definition 2.2 (Canonical real representations and cost functions)**.** A *component* (`Component`) is a pair $(\lambda, P) \in \mathbb{C} \times \mathbb{C}[X]$, evaluating at $s \in \mathbb{R}$ as $\operatorname{Re}(P(s)e^{\lambda s})$ of degree $k = \deg P$ (multiplicity $k+1$). It is *well-formed* if $P \neq 0$, $\operatorname{Im}\lambda \geq 0$, and $\operatorname{Im}\lambda = 0 \implies P \in \mathbb{R}[X]$. A finite list $\mathcal{R} = [(\lambda_1, P_1), \dots, (\lambda_m, P_m)]$ is a *representation* (`IsRepresentation`) if every (λ_u, P_u) is well-formed and the rates $\lambda_1, \dots, \lambda_m$ are pairwise distinct; its radial profile is $\operatorname{profile}(\mathcal{R}, s) = \sum_{u=1}^{m} \operatorname{Re}(P_u(s)e^{\lambda_u s})$. For $d \geq 2$, define the exact d-way component cost $c_d(\lambda, P)$ and multi-Gaussian border-rank cost $b_d(\lambda, P)$ by

$$c_d(\lambda, P) = \begin{cases} (d-1)\deg P + 1, & \operatorname{Im}\lambda = 0, \\ 2(d-1)\deg P + d, & \operatorname{Im}\lambda > 0, \end{cases} \qquad b_d(\lambda, P) = \begin{cases} \deg P + 1, & \operatorname{Im}\lambda = 0, \\ d(\deg P + 1), & \operatorname{Im}\lambda > 0, \end{cases} \tag{5}$$

with total costs $C_d(\mathcal{R}) = \sum_{u=1}^{m} c_d(\lambda_u, P_u)$ (`multiTotalCost`) and $B_d(\mathcal{R}) = \sum_{u=1}^{m} b_d(\lambda_u, P_u)$ (`borderTotalCost`). When every $\deg P_u = 0$ (*multi-Gaussian* representation with n_{re} real and n_{cx} nonreal rates), $C_d(\mathcal{R}) = B_d(\mathcal{R}) = n_{\mathrm{re}} + d\, n_{\mathrm{cx}}$.

Why are the conditions $\operatorname{Im}\lambda \geq 0$ and $(\operatorname{Im}\lambda = 0 \implies P \in \mathbb{R}[X])$ canonical? Because $\operatorname{Re}(\overline{P}(s)e^{\bar{\lambda}s}) = \operatorname{Re}(P(s)e^{\lambda s})$ and $\operatorname{Re}(iae^{\mu s}) = 0$ for $a, \mu \in \mathbb{R}$, unrestricted pairs over-parameterize real profiles. Under Definition 2.2, each real rate ($\operatorname{Im}\lambda = 0$) contributes a single term $P(s)e^{\lambda s}$ over $\mathbb{C}$, while each upper-half-plane rate ($\operatorname{Im}\lambda > 0$) expands via $\operatorname{Re} z = \frac{1}{2}(z + \bar{z})$ to the conjugate pair $\frac{1}{2}P(s)e^{\lambda s} + \frac{1}{2}\overline{P}(s)e^{\bar{\lambda}s}$ at distinct rates $\lambda \neq \bar{\lambda}$. Since exponential polynomials at distinct complex rates are $\mathbb{C}$-linearly independent on $(0, \infty)$, every continuous real profile of finite rank determines its unordered set of well-formed components uniquely. Table 2 summarizes all symbols and their Lean 4 counterparts.

2.2 Statement of main theorems and algebraic identities

Theorem 2.3 (d-D Finite-Rank Classification; Lean: `separableRank_radial_lt_top_iff_exists`)**.** *Let $d \geq 2$ and let $\varphi : \mathbb{R} \to \mathbb{R}$ be continuous on $[0, \infty)$. Then $\operatorname{srank}_{\mathbb{R}}\big(x \in \mathbb{R}^d \mapsto \varphi(\sum_{j=0}^{d-1} x_j^2)\big) < \infty$ if and only if there exists a representation $\mathcal{R}$ such that $\varphi(s) = \operatorname{profile}(\mathcal{R}, s)$ for all $s \geq 0$.*

Theorem 2.4 (d-D Complex Radial Rank; Lean: `separableRank_radial_complex_eq_iff`)**.** *Let $d \geq 2$ and let $\psi : \mathbb{R} \to \mathbb{C}$ be continuous on $[0, \infty)$. For every $R \in \mathbb{N}$,*

$$\operatorname{srank}_{\mathbb{C}}\left(x \in \mathbb{R}^d \mapsto \psi\left(\sum_{j=0}^{d-1} x_j^2\right)\right) = R$$

if and only if there exist a finite set $S \subset \mathbb{C}$ and nonzero polynomials $(P_\mu)_{\mu \in S} \subset \mathbb{C}[X]$ such that $\psi(s) = \sum_{\mu \in S} P_\mu(s)e^{\mu s}$ for all $s \geq 0$ and $\sum_{\mu \in S}\big((d-1)\deg P_\mu + 1\big) = R$.

Theorem 2.5 (Exact d-D Real Radial Rank; Lean: `separableRank_radial_eq_iff`)**.** *Let $d \geq 2$ and let $\varphi : \mathbb{R} \to \mathbb{R}$ be continuous on $[0, \infty)$. For every $R \in \mathbb{N}$,*

$$\operatorname{srank}_{\mathbb{R}}\left(x \in \mathbb{R}^d \mapsto \varphi\left(\sum_{j=0}^{d-1} x_j^2\right)\right) = R$$

if and only if there exists a representation $\mathcal{R}$ such that $\varphi(s) = \operatorname{profile}(\mathcal{R}, s)$ for all $s \geq 0$ and $C_d(\mathcal{R}) = R$. In particular, for a multi-Gaussian representation with n_{re} real and n_{cx} nonreal components, $\operatorname{srank}_{\mathbb{R}} = n_{\mathrm{re}} + d\, n_{\mathrm{cx}}$, and for $d = 2$, $\operatorname{srank}_{\mathbb{R}} = C_2(\mathcal{R}) = \operatorname{srank}_{\mathbb{C}}(x \mapsto \varphi_{\mathbb{C}}(x_0^2 + x_1^2))$.

Theorem 2.6 (2-D Minimal Differential Annihilator; Lean: `exists_minimal_differential_annihilator`)**.** *Let $\varphi : \mathbb{R} \to \mathbb{R}$ be continuous on $[0, \infty)$ with $\operatorname{srank}_{\mathbb{R}}(x \in \mathbb{R}^2 \mapsto \varphi(x_0^2 + x_1^2)) = R < \infty$. For any $q(X) = \sum_{k=0}^{\deg q} q_k X^k \in \mathbb{C}[X]$ and $g : \mathbb{R} \to \mathbb{C}$, define the half-line differential operator (`differentialOperator`)*

$$\big(q(D)g\big)(t) := \sum_{k=0}^{\deg q} q_k\, D^k_{[0,\infty)} g(t),$$

where $D^0_{[0,\infty)} g = g$ and $D^k_{[0,\infty)} g(t)$ (`iteratedDerivWithin k g (Set.Ici 0) t`, $k \geq 1$) is Mathlib's totalized k-th derivative within $[0, \infty)$ (returning 0 where a within-derivative fails to exist, and agreeing for $g \in C^\infty([0, \infty))$ with the classical interior derivative at $t > 0$ and the right-hand limit $\lim_{h \to 0^+} h^{-1}(D^{k-1}_{[0,\infty)} g(h) - D^{k-1}_{[0,\infty)} g(0))$ at $t = 0$). Then $\varphi_{\mathbb{C}}|_{[0,\infty)} \in C^\infty([0, \infty))$ (so every $D^k_{[0,\infty)} \varphi_{\mathbb{C}}(t)$ is a genuine classical derivative), and there exists a unique monic polynomial $p_{\min} \in \mathbb{C}[X]$ of degree R such that for every $q \in \mathbb{C}[X]$, $\big(q(D)\varphi_{\mathbb{C}}\big)(t) = 0$ for all $t \geq 0$ if and only if $p_{\min} \mid q$.

Remark 2.7 (Automatic smoothness, domain of D, and identity with the generator's characteristic polynomial)**.** Theorem 2.6 assumes *only* continuity of φ on $[0,\infty)$ and $\operatorname{srank}_{\mathbb{R}} = R < \infty$, not differentiability: Section 3.1 shows by integral smoothing of the shift semigroup that $\varphi_{\mathbb{C}}$ agrees on $[0,\infty)$ with an analytic exponential polynomial $\sum_{\mu\in S} P_\mu(t)e^{\mu t}$, so every classical right-hand derivative $D^k_{[0,\infty)}\varphi_{\mathbb{C}}(t)$ ($t \ge 0$, $k \ge 0$) exists automatically! (Conversely, any continuous profile on $[0,\infty)$ that fails to be C^∞—such as $|s-1|$—has $\operatorname{srank}_{\mathbb{R}} = \infty$. In Lean, `derivWithin` is totalized to return 0 where a derivative does not exist, so restricting Theorem 2.6 to finite-rank continuous profiles ensures that $D^k_{[0,\infty)}\varphi_{\mathbb{C}}$ always coincides with the classical within-derivative.) Uniqueness of $p_{\min}$ is immediate because any two monic polynomials dividing each other are equal; moreover, in terms of the shift generator $A \in \mathcal{L}(\mathcal{W}(\varphi_{\mathbb{C}}))$ of Section 3.1, $\deg p_{\min} = R = \dim_{\mathbb{C}} \mathcal{W}(\varphi_{\mathbb{C}}) = \deg \chi_A$ together with $p_{\min} \mid \chi_A$ implies $p_{\min}(X) = \chi_A(X) = \mu_A(X)$ ($\varphi_{\mathbb{C}}|_{\mathbb{R}_{\ge 0}}$ is a cyclic vector for A).

In $d = 2$ (matrices), a pointwise limit of rank-$\le B$ kernels always has rank at most B because every $(B+1)\times(B+1)$ evaluation minor vanishes (compare Laird's closedness characterization of finite-dimensional translate spans under compact-open convergence [45]). In $d \ge 3$ dimensions, however, a kernel of exact rank R can be the limit of kernels of strictly smaller rank $B < R$ whose weights grow as $O(1/h)$ and cancel as $h \to 0$—just as $s_0 + s_1 + s_2 = \lim_{h\to 0} h^{-1}\big((1+hs_0)(1+hs_1)(1+hs_2) - 1\big)$ approximates a rank-3 sum by a difference of *two* 3-way products [8, 47]. The least $B \in \mathbb{N}$ for which K is a uniform limit on every compact set $C \subset \mathbb{R}^d$ of kernels of rank $\le B$ is the *compact-open border rank* of K:

Theorem 2.8 (General d-D Multi-Gaussian Border-Rank Bound and Gap)**.** *Let $d \ge 2$ and let $\mathcal{R} = [(\lambda_1, P_1), \dots, (\lambda_m, P_m)]$ be any finite list of well-formed components. Formalized as* `multiTotalCost_eq_borderTotalCost_add` *and* `exists_rank_le_borderTotalCost_tendstoUniformlyOn_radial_profile`*:*

1. Exact gap identity: *Both real and nonreal components satisfy $c_d(\lambda_u, P_u) = b_d(\lambda_u, P_u) + (d-2)\deg P_u$, giving*
$$C_d(\mathcal{R}) = B_d(\mathcal{R}) + (d-2)\sum_{u=1}^{m} \deg P_u. \tag{6}$$

2. Uniform multi-Gaussian approximation on compact sets: *There exists an explicit family of multi-Gaussian kernels $F_h : \mathbb{R}^d \to \mathbb{R}$ ($h \in \mathbb{R}\setminus\{0\}$, extended to $h = 0$ in Lean via $0^{-1} = 0$) satisfying $\operatorname{srank}_{\mathbb{R}}(F_h) \le B_d(\mathcal{R})$ for all h and $F_h(x) \to \operatorname{profile}(\mathcal{R}, \|x\|^2)$ uniformly on every compact subset $C \subset \mathbb{R}^d$ as $h \to 0$ ($h \ne 0$). Consequently, whenever $\mathcal{R}$ is a representation in dimension $d \ge 3$ with $\sum_u \deg P_u \ge 1$, its compact-open real border rank is strictly less than its exact real rank $C_d(\mathcal{R})$ by at least $(d-2)\sum_u \deg P_u \ge d-2$. In particular, for $K_d(x) = \|x\|^2 \cos(\|x\|^2)$ ($\mathcal{R} = [(i, X)]$), $\operatorname{srank}_{\mathbb{C}}((K_d)_{\mathbb{C}}) = 2d$, $\operatorname{srank}_{\mathbb{R}}(K_d) = 3d-2$, and $\operatorname{srank}_{\mathbb{R}}(F_h) \le 2d$.*

The upper bounds for nonreal components in Theorems 2.5 and 2.8 rest on two root-of-unity identities of independent algebraic interest (throughout, $[X^k]F(X)$ denotes the coefficient of X^k in a polynomial $F \in \mathbb{C}[X]$):

Proposition 2.9 (Scalar alternating $2d$-th root-of-unity identity; Lean: `re_prod_eq_sum_re`)**.** *For every integer $d \ge 1$ and all $z_0, \dots, z_{d-1} \in \mathbb{C}$,*
$$\operatorname{Re}\left(\prod_{j=0}^{d-1} z_j\right) = \frac{2^{d-1}}{d}\sum_{r=0}^{d-1}(-1)^r \prod_{j=0}^{d-1} \operatorname{Re}\left(z_j e^{-i\pi r/d}\right). \tag{7}$$

Proposition 2.10 (Universal polynomial-product root-of-unity identity; Lean: `re_coeff_prod_eq_sum_re`)**.** *Let $d \geq 2$, $k \geq 0$, and $M^{\mathbb{R}}_{d,k} = 2(d-1)k + d$. For any polynomials $U_0, \ldots, U_{d-1} \in \mathbb{C}[X]$ with $\deg U_j \leq k$,*

$$\operatorname{Re}\left([X^k] \prod_{j=0}^{d-1} U_j(X) \right) = \frac{2^{d-1}}{M^{\mathbb{R}}_{d,k}} \sum_{r=0}^{M^{\mathbb{R}}_{d,k}-1} (-1)^r \prod_{j=0}^{d-1} \operatorname{Re}\left(U_j\left(e^{\frac{2\pi i r}{M^{\mathbb{R}}_{d,k}}} \right) e^{-\frac{i\pi r(2k+1)}{M^{\mathbb{R}}_{d,k}}} \right). \tag{8}$$

When $k = 0$, $M^{\mathbb{R}}_{d,0} = d$, each $U_j(X) = z_j$ is a constant polynomial, $[X^0] \prod_j z_j = \prod_j z_j$, and the phase $e^{-i\pi r(2\cdot 0+1)/d} = e^{-i\pi r/d}$ reduces (8) identically to (7).

Example 2.11 (Concrete $d = 3, k = 0$ complex Gaussian: 3 real products vs. 4 naive products)**.** Let $d = 3$ and write $z_j = a_j + ib_j \in \mathbb{C}$ $(a_j, b_j \in \mathbb{R})$ for $z_j(x_j) = \beta^{1/3} e^{(-\alpha+i\omega)x_j^2}$. Expanding $\operatorname{Re}(z_0 z_1 z_2)$ directly into real and imaginary parts requires $2^{3-1} = 4$ real 3-way products:

$$\operatorname{Re}(z_0 z_1 z_2) = a_0 a_1 a_2 - a_0 b_1 b_2 - b_0 a_1 b_2 - b_0 b_1 a_2.$$

Instead, Proposition 2.9 at $d = 3$ evaluates $u_{j,r} = \operatorname{Re}(z_j e^{-i\pi r/3}) = a_j \cos(\frac{\pi r}{3}) + b_j \sin(\frac{\pi r}{3})$ at the three angles $0, \frac{\pi}{3}, \frac{2\pi}{3}$ $(u_{j,0} = a_j,\ u_{j,1} = \frac{1}{2}a_j + \frac{\sqrt{3}}{2}b_j,\ u_{j,2} = -\frac{1}{2}a_j + \frac{\sqrt{3}}{2}b_j)$:

$$\operatorname{Re}(z_0 z_1 z_2) = \frac{4}{3}\Big(u_{0,0}u_{1,0}u_{2,0} - u_{0,1}u_{1,1}u_{2,1} + u_{0,2}u_{1,2}u_{2,2} \Big).$$

Expanding the right-hand side confirms that all $a_0a_1b_2$ and $b_0b_1b_2$ terms cancel via $-\frac{\sqrt{3}}{8} + \frac{\sqrt{3}}{8} = 0$ and $-\frac{3\sqrt{3}}{8} + \frac{3\sqrt{3}}{8} = 0$, while $a_0a_1a_2$ keeps $1 - \frac{1}{8} - \frac{1}{8} = \frac{3}{4}$, leaving $\frac{4}{3}(\frac{3}{4}a_0a_1a_2 - \frac{3}{4}(a_0b_1b_2 + b_0a_1b_2 + b_0b_1a_2)) = \operatorname{Re}(z_0z_1z_2)$ using *three* symmetric real 3-way products rather than four, and Theorem 2.5 proves that two never suffice when $\omega > 0$. Algebraically, pairing $(z_0, z_1) \mapsto z_0 z_1$ against $\bar{z}_2 \mapsto \operatorname{Re}((z_0z_1)z_2)$ identifies this $2 \times 2 \times 2$ real tensor with bilinear multiplication in $\mathbb{C} \cong \mathbb{R}[X]/(X^2+1)$, whose real rank is 3 by Gauss/Karatsuba complex multiplication [76].

Example 2.12 (Concrete $d = 3, k = 1$ polynomial oscillation: 7 real products via $M^{\mathbb{R}}_{3,1} = 7$ vs. 9 naive)**.** Consider $K_3(x) = (x_0^2 + x_1^2 + x_2^2)\cos(x_0^2 + x_1^2 + x_2^2)$ $(\lambda = i,\ P(s) = s,\ k = 1)$. Splitting $x_0^2 \cos\|x\|^2 + x_1^2 \cos\|x\|^2 + x_2^2 \cos\|x\|^2$ into three shifted Gaussians and applying Example 2.11 to each costs $3 + 3 + 3 = 9$ real 3-way products. Instead, put $U_j(X) = e^{ix_j^2}(1 + x_j^2 X)$ $(j = 0, 1, 2)$, so $[X^1](U_0U_1U_2) = (x_0^2 + x_1^2 + x_2^2)e^{i\|x\|^2}$. Here $M^{\mathbb{R}}_{3,1} = 2(3-1)(1) + 3 = 7$ and $2k + 1 = 3$; evaluating $\operatorname{Re}\big(U_j(e^{2\pi ir/7})e^{-3\pi ir/7}\big)$ gives the seven real univariate functions

$$K_3(x) = \frac{4}{7}\sum_{r=0}^{6}(-1)^r f_r(x_0)\, f_r(x_1)\, f_r(x_2), \qquad f_r(u) := \cos\left(u^2 - \frac{3\pi r}{7}\right) + u^2 \cos\left(u^2 - \frac{\pi r}{7}\right).$$

Why must the node count be 7 with alternating signs $(-1)^r = e^{7\pi ir/7}$? Writing $\cos\theta = \frac{1}{2}(e^{-i\theta} + e^{i\theta})$ in $f_r(x_0)f_r(x_1)f_r(x_2)$ produces harmonics $e^{2\pi irK/7}$ where $m \in \{0, 1, 2, 3\}$ factors pick the $-i$ branch and $q_j \in \{0, 1\}$ picks 1 vs. x_j^2, giving $K = 3m + \sum_{j \notin S} q_j - \sum_{j \in S} q_j - 1$. For mixed branches $m = 1$ $(K \in \{1, \ldots, 4\})$ and $m = 2$ $(K \in \{3, \ldots, 6\})$, 7 is the smallest modulus keeping every mixed harmonic inside $\{1, \ldots, M-1\}$ so $\sum_{r=0}^{6} e^{2\pi irK/7} = 0$, while on pure branches $m = 0$ $(K = q_0 + q_1 + q_2 - 1)$ and $m = 3$ $(K = 8 - q_0 - q_1 - q_2)$ only $q_0 + q_1 + q_2 = 1$ is divisible by 7, isolating $K_3(x)$ in 7 real products (proved optimal in Section 3.3).

2.3 Intuitive guide: what $c_d(\lambda, P)$ counts and why the $d-2$ penalty arises

For readers primarily interested in filter design or scientific computing rather than commutative algebra, four plain-language rules explain every number in Theorems 2.3–2.8 (illustrated in Table 3):

Table 3: Component-cost comparison across $d \in \{2,3,4\}$ and degrees $k = \deg P \in \{0,1\}$: exact complex rank, exact real rank $c_d(\lambda, P)$ (each term costing d one-dimensional passes), naive real expansion (grouping p_0 into mode 0 to form d complex products at $k=1$ and expanding each $\mathrm{Re}(z_0 \cdots z_{d-1})$ into 2^{d-1} real products, giving $d\,2^{d-1}$), and multi-Gaussian border-rank bound $b_d(\lambda, P)$.

Single component type	Profile $\varphi(s)$ ($s = \|x\|^2$)	Dim. d	Complex rank	Exact real c_d	Naive real	Border bound b_d
Real Gaussian ($k=0$)	$a\,e^{-\alpha s}$	$2,3,4$	$1,1,1$	$1,1,1$	$1,1,1$	$1,1,1$
Real Laguerre ($k=1$)	$(p_0 + p_1 s)e^{-\alpha s}$	$2,3,4$	$2,3,4$	$2,3,4$	$2,3,4$	$2,2,2$
Complex Gaussian ($k=0$)	$\mathrm{Re}(\beta\, e^{(-\alpha+i\omega)s})$ ($\omega > 0$)	$2,3,4$	$2,2,2$	$2,3,4$	$2,4,8$	$2,3,4$
Complex poly. ($k=1$)	$\mathrm{Re}((p_0 + p_1 s)e^{(-\alpha+i\omega)s})$	$2,3,4$	$4,6,8$	$4,7,10$	$4,12,32$	$4,6,8$

1. **What $c_d(\lambda, P)$ counts computationally:** A single component (λ, P) is one radially symmetric building block (such as u_1 or u_2 in Figure 1a), whereas $c_d(\lambda, P)$ counts how many rank-1 real product channels $\prod_{j=0}^{d-1} f_{n,j}(x_j)$ (the rendering terms t_n in Figure 1b) must be summed to evaluate that component over $\mathbb{R}$. On a d-dimensional grid of width W, rank $R = C_d(\mathcal{R}) = \sum_u c_d(\lambda_u, P_u)$ executes dR one-dimensional passes (dRW multiply–adds per voxel).

2. **Why a real rate costs $(d-1)k+1$:** Expanding $P(s_0 + \cdots + s_{d-1})$ of degree k places a degree-k polynomial on each of the last $d-1$ coordinates; their product has degree $(d-1)k$, which has $(d-1)k+1$ independent coefficients (1 when $k=0$; d when $k=1$, matching $s_0 + \cdots + s_{d-1}$) and can be reconstructed from $(d-1)k+1$ real evaluation points.

3. **Where the constant $d-2$ real-over-complex penalty comes from:** Over $\mathbb{C}$, a nonreal component $\mathrm{Re}(P(s)e^{\lambda s}) = \frac{1}{2}P(s)e^{\lambda s} + \frac{1}{2}\overline{P}(s)e^{\bar{\lambda} s}$ splits into two independent complex rates of cost $(d-1)k+1$ each, totaling $2(d-1)k+2$ (2 complex products when $k=0$, in *every* dimension d). Over $\mathbb{R}$, however, even at $k=0$ ($\mathrm{Re}(z_0 \cdots z_{d-1})$ with $z_j \in \mathbb{C} \cong \mathbb{R}^2$), every real linear functional $\ell : \mathbb{C} \to \mathbb{R}$ has a 1-dimensional nullspace: choosing $z_j \neq 0$ in that nullspace kills one real product term per coordinate $j = 0, \ldots, d-2$ while leaving the remaining product nonzero, forcing at least d real products (2 in 2-D, 3 in 3-D, 4 in 4-D)—a base penalty of $d-2$ over the 2 complex products! Each polynomial degree increment $k \mapsto k+1$ then adds $2(d-1)$ real degrees of freedom on modes $0, \ldots, d-2$ over *both* $\mathbb{C}$ and $\mathbb{R}$, so $c_d(\lambda, P) = d + 2(d-1)k$ and the complex pair cost $2 + 2(d-1)k$ grow at the same rate $2(d-1)k$, keeping their difference locked at $d-2$ for every $k \geq 0$.

4. **Why the multi-Gaussian border-rank bound drops by $(d-2)k$:** Approximating $se^{\lambda s} \approx \frac{e^{(\lambda+h)s} - e^{\lambda s}}{h}$ replaces one degree-1 polynomial factor by a difference of *two* pure Gaussians ($k=0$): in $d \geq 3$, two complex Gaussians cost $2d$ real products whereas exact $k=1$ costs $2(d-1)+d = 3d-2$, lowering the multi-Gaussian border-rank upper bound $B_d(\mathcal{R})$ by $d-2$ products per polynomial degree at the price of $O(1/h)$ cancelling weights as $h \to 0$.

Remark 2.13 (Unified commutative-algebra invariant and comparison with symmetric Waring rank)**.** Write $\mathcal{A}_\varphi = \mathbb{R}[X]/(p_{\min,\mathbb{R}}(X))$ for the finite-dimensional commutative polynomial quotient algebra associated to the minimal real differential annihilator $p_{\min,\mathbb{R}}(D)\varphi = 0$ (isomorphic to $\mathcal{W}(\varphi)$ as an $\mathbb{R}[D]$-module, with multiplication induced by composition of D rather than pointwise product of profiles). Over $\mathbb{R}$, $p_{\min,\mathbb{R}}(X)$ factors into $m_{\mathrm{irr}} = n_{\mathrm{re}} + n_{\mathrm{cx}}$ distinct powers of irreducible real polynomials (n_{re} linear factors $(X - \lambda_u)^{k_u+1}$ and n_{cx} irreducible real quadratics $((X+\alpha_u)^2 + \omega_u^2)^{k_u+1}$; equivalently, m_{irr} maximal ideals of $\mathcal{A}_\varphi$), while $\dim_{\mathbb{R}} \mathcal{A}_\varphi = \deg p_{\min,\mathbb{R}} = C_2(\mathcal{R})$. Checking $(d-1)(k+1) - (d-2)\cdot 1 = (d-1)k + 1$ and $(d-1)\cdot 2(k+1) - (d-2)\cdot 1 = 2(d-1)k + d$ shows that Theorem 2.5 takes one unified form

across all $d \geq 2$:

$$\operatorname{srank}_{\mathbb{R}}(K) = C_d(\mathcal{R}) = (d-1)\dim_{\mathbb{R}} \mathcal{A}_\varphi - (d-2)\, m_{\mathrm{irr}} \qquad (d \geq 2), \tag{9}$$

and the identical formula holds over $\mathbb{C}$ with $m_{\mathrm{irr}} = |S|$ linear factors. At $d = 2$ (1-fold identity pairing), (9) reduces to $\dim_{\mathbb{K}} \mathcal{A}$; at $d = 3$ (2-fold bilinear multiplication), it coincides with the Winograd–Fiduccia–Zalcstein–Alder–Strassen complexity $2\dim_{\mathbb{K}} \mathcal{A} - m_{\mathrm{irr}}$ [76, 23, 1, 15]; and for $d \geq 4$ ($(d-1)$-fold multiplication), it matches the multilinear Alder–Strassen lower bound of Bläser, Mayer, and Shringi [10] with equality. Notice the contrast with symmetric Waring rank: for a real binary form $F(u, v) = \mathrm{Re}((u + iv)^D)$ ($D \geq 2$), complex Waring rank is 2 whereas real symmetric Waring rank equals D [18, 63, 11]—a gap $D - 2$ that grows with degree. In (9), complexifying $\mathbb{R} \hookrightarrow \mathbb{C}$ splits each irreducible real quadratic into two conjugate linear factors (m_{irr} increases by 1 per nonreal pair while $\dim \mathcal{A}$ is unchanged), producing a *degree-independent* real-over-complex penalty of $(d-2)(2-1) = d-2$ per conjugate pair because coordinatewise decompositions (1) break mode symmetry by placing $H_P(x_0^2; X)$ on mode 0 and $E_k(x_j^2; X)$ on modes $j \geq 1$.

3 Mathematical Proofs and Core Constructions

Radial $K(x) = f(\|x\|^2)$ on $\mathbb{R}^d$ $\xleftarrow[\text{surjection } s_j = x_j^2]{\text{Eq. (3)}}$ Hankel $(s_0, \ldots, s_{d-1}) \mapsto f\left(\sum_{j=0}^{d-1} s_j\right)$ on $\mathbb{R}^d_{\geq 0}$

Constructive upper bounds ($\leq R$)
- $\mathbb{C}$: $M^{\mathbb{C}}_{d,k} = (d-1)k+1$ roots of unity (Sec. 3.2)
- $\mathbb{R}$, $\mathrm{Im}\,\lambda{=}0$: Lagrange at $0, \ldots, M^{\mathbb{C}}_{d,k}-1$ (Sec. 3.2)
- $\mathbb{R}$, $\mathrm{Im}\,\lambda{>}0$: $M^{\mathbb{R}}_{d,k} = 2(d-1)k+d$ roots (Prop. 2.10)

Necessity & multilinear lower bounds ($\geq R$)
- $d{=}2$ slice $\Rightarrow \dim_{\mathbb{C}} \mathcal{W}(\psi) < \infty \Rightarrow T_t = e^{tA}$ (Sec. 3.1)
- Dual span of $v(\mathbb{R}_{\geq 0}) \Rightarrow$ linear $\ell_{n,j} \in V^*$ (Sec. 3.3)
- Alder–Strassen peeling + $+1$ shift (Sec. 3.3)

3.1 Necessity and the two-dimensional core

In two coordinates over a field $\mathbb{K}$, let $\mathrm{rowSpan}(K) \leq (X \to \mathbb{K})$ be the $\mathbb{K}$-linear span of the row sections $\{x \mapsto K(x, y) : y \in X\}$. Then $\operatorname{srank}_{\mathbb{K}}(K) = \dim_{\mathbb{K}} \mathrm{rowSpan}(K) \in \mathbb{N} \cup \{\infty\}$: if $K(x, y) = \sum_{n=0}^{R-1} f_n(x) g_n(y)$, every row $x \mapsto K(x, y)$ is a linear combination of $f_0, \ldots, f_{R-1}$ so $\dim \mathrm{rowSpan}(K) \leq R$; conversely, if $b_0(x), \ldots, b_{R-1}(x)$ is a basis of $\mathrm{rowSpan}(K)$, expanding each row $x \mapsto K(x, y) = \sum_{n=0}^{R-1} c_n(y) b_n(x)$ directly gives a rank-R sum of products. For the two-variable Hankel kernel $(s_0, s_1) \mapsto f(s_0 + s_1)$ on $\mathbb{R}^2_{\geq 0}$, the row span is the *translate span*

$$\mathcal{W}(f) = \operatorname{span}_{\mathbb{K}}\{u \in \mathbb{R}_{\geq 0} \mapsto f(u + t) : t \in \mathbb{R}_{\geq 0}\} \leq (\mathbb{R}_{\geq 0} \to \mathbb{K}).$$

Scalar extension $\mathbb{R} \hookrightarrow \mathbb{C}$ satisfies $\operatorname{srank}_{\mathbb{C}}(K_{\mathbb{C}}) \leq \operatorname{srank}_{\mathbb{R}}(K)$ in every dimension d. In $d = 2$, equality $\operatorname{srank}_{\mathbb{C}}(K_{\mathbb{C}}) = \operatorname{srank}_{\mathbb{R}}(K)$ always holds: if real functions $b_0, \ldots, b_{R-1} : X \to \mathbb{R}$ are $\mathbb{R}$-linearly independent, any complex relation $\sum_n (\alpha_n + i\beta_n) b_n(x) = 0$ forces $\sum_n \alpha_n b_n(x) = 0$ and $\sum_n \beta_n b_n(x) = 0$ separately, so $\dim_{\mathbb{C}} \mathrm{rowSpan}(K_{\mathbb{C}}) = \dim_{\mathbb{R}} \mathrm{rowSpan}(K)$.

For any $d \geq 2$, if $\operatorname{srank}_{\mathbb{R}}(x \in \mathbb{R}^d \mapsto \varphi(\|x\|^2)) < \infty$, the slice inequality (4) and 2-D complexification give $\dim_{\mathbb{C}} \mathcal{W}(\varphi_{\mathbb{C}}) < \infty$. Necessity and the 2-D annihilator theorem then follow from a single basis-free operator construction on $\mathcal{W}(\psi)$ for any continuous $\psi : \mathbb{R} \to \mathbb{C}$ with $\dim_{\mathbb{C}} \mathcal{W}(\psi) = n < \infty$:

1. Equip the algebraic subspace $\mathcal{W}(\psi) \leq (\mathbb{R}_{\geq 0} \to \mathbb{C})$—not a space of bounded continuous functions, since ψ may grow polynomially or exponentially on $[0, \infty)$—with the Hausdorff product topology of pointwise convergence inherited from $\mathbb{R}_{\geq 0} \to \mathbb{C}$. Because $\dim_{\mathbb{C}} \mathcal{W}(\psi) = n < \infty$ and all Hausdorff vector topologies on an n-dimensional $\mathbb{C}$-vector space coincide (`isNormableSpace_of_finiteDimensional`), $\mathcal{W}(\psi)$ admits a compatible norm whose topology equals pointwise convergence and is automatically complete (`FiniteDimensional.complete`), making $\mathcal{W}(\psi)$ a complex Banach space without any growth or smoothness hypothesis on ψ.

2. Non-negative shifts $(T_t g)(u) = g(u+t)$ $(t \geq 0)$ map $\mathcal{W}(\psi)$ into itself because shifting a translate $u \mapsto \psi(u+s)$ gives $u \mapsto \psi(u+(s+t)) \in \mathcal{W}(\psi)$, requiring no derivatives of ψ. Continuity of ψ on $[0,\infty)$ implies that $t \mapsto (T_t g)(u)$ is continuous at each point $u \geq 0$ on every spanning translate, hence norm-continuous on $\mathcal{W}(\psi)$, and therefore $t \mapsto T_t \in \mathcal{L}(\mathcal{W}(\psi))$ is continuous in operator norm on $[0,\infty)$ (evaluating a linear map on a finite basis of $\mathcal{W}(\psi)$).

3. By the Banach-algebra semigroup theorem (Section 4, which manufactures differentiability via integral smoothing $\int_t^{t+\delta} T_u\, du$; cf. Anselone and Korevaar's matrix-semigroup proof on $\mathbb{R}$ [3, p. 750]), there exists a generator $A \in \mathcal{L}(\mathcal{W}(\psi))$ with $T_t = \exp(tA)$ for all $t \geq 0$, so $\psi(t) = (T_t \psi|_{\mathbb{R}_{\geq 0}})(0) = \mathrm{eval}_0(\exp(tA)\psi|_{\mathbb{R}_{\geq 0}})$ is automatically C^∞ on $[0,\infty)$.

4. Over the algebraically closed field $\mathbb{C}$, $\mathcal{W}(\psi) = \bigoplus_{\mu \in S} \ker((A-\mu I)^n)$ decomposes into generalized eigenspaces. On a summand where $(A-\mu I)^n v = 0$, writing $tA = \mu t I + t(A-\mu I)$ as a sum of commuting operators gives $\exp(tA)v = e^{\mu t} \sum_{q=0}^{n-1} \frac{t^q}{q!}(A-\mu I)^q v$, because all powers $q \geq n$ vanish. Evaluating at 0 turns each vector $(A-\mu I)^q v$ into a complex scalar, yielding $\psi(t) = \sum_{\mu \in S} P_\mu(t) e^{\mu t}$ on $[0,\infty)$. When $\psi = \varphi_{\mathbb{C}}$ for a real profile φ, averaging $\varphi(t) = \frac{1}{2}(\varphi_{\mathbb{C}}(t) + \overline{\varphi_{\mathbb{C}}(t)})$ and grouping conjugate rates $(\mu, \bar{\mu})$ produces a canonical real representation $\mathcal{R}$.

Finally, differentiating $t \mapsto \exp(tA)\psi|_{\mathbb{R}_{\geq 0}}$ q times on $[0,\infty)$ brings down A^q, so $q(D)\psi(t) = \mathrm{eval}_0\big(q(A)\exp(tA)\psi|_{\mathbb{R}_{\geq 0}}\big)$ for every polynomial $q \in \mathbb{C}[X]$. In $d=2$, the Cayley–Hamilton theorem ($q = \chi_A \implies \chi_A(A) = 0$) immediately implies that the monic characteristic polynomial $\chi_A(X)$ of degree $\dim_{\mathbb{C}} \mathcal{W}(\psi)$ satisfies $\chi_A(D)\psi = 0$ on $[0,\infty)$. Conversely, applying $(D-\mu)^{\deg P_\mu + 1}$ kills $P_\mu(t)e^{\mu t}$ while shifting every other rate $\nu \neq \mu$ by an invertible factor $(\nu-\mu)^{\deg P_\mu + 1}$ on leading terms, so $q(D)\big(\sum_{\mu \in S} P_\mu(t)e^{\mu t}\big) = 0$ on $[0,\infty)$ ($P_\mu \neq 0$) if and only if $p_{\min}(X) = \prod_{\mu \in S}(X-\mu)^{\deg P_\mu + 1}$ divides $q(X)$. Hence $p_{\min} \mid \chi_A$, giving $\sum_{\mu \in S}(\deg P_\mu + 1) = \deg p_{\min} \leq \deg \chi_A = \dim_{\mathbb{C}} \mathcal{W}(\psi)$, matched by Taylor expansion of $P_\mu(s_0+s_1)e^{\mu(s_0+s_1)}$ to prove Theorem 2.6.

3.2 Constructive multi-way upper bounds via Hasse polynomials and roots of unity

Warm-up ($d=3, k=1$): extracting a degree-1 radial polynomial as an $[X^1]$ coefficient. Consider $P(s) = p_0 + p_1 s$ on $s = s_0 + s_1 + s_2$. Notice that if we put $U_0(X) = p_1 + (p_0 + p_1 s_0)X$, $U_1(X) = 1 + s_1 X$, and $U_2(X) = 1 + s_2 X$, then the coefficient of X^1 in $U_0(X)U_1(X)U_2(X)$ is

$$[X^1]\big((p_1+(p_0+p_1s_0)X)(1+s_1X)(1+s_2X)\big) = (p_0+p_1s_0)\cdot 1\cdot 1 + p_1 s_1 \cdot 1 + p_1 \cdot 1 \cdot s_2 = P(s_0+s_1+s_2),$$

a product of three univariate polynomials in X whose j-th factor depends *only on* s_j! Since $\deg(U_0U_1U_2) \leq 3$, over $\mathbb{C}$ averaging $\zeta^{-r}U_0(\zeta^r)U_1(\zeta^r)U_2(\zeta^r)$ over the $M_{3,1}^{\mathbb{C}} = 3$ cube roots of unity ($\zeta = e^{2\pi i/3}$) isolates the X^1 term (because X^0, X^2, X^3 shift to X^{-1}, X^1, X^2, none of which is a multiple of 3!), whereas over $\mathbb{R}$ for a nonreal rate $e^{\lambda(s_0+s_1+s_2)}$ ($\mathrm{Im}\,\lambda > 0$), Proposition 2.10 extracts $\mathrm{Re}([X^1]\prod_{j=0}^2 e^{\lambda s_j} U_j(X))$ using $M_{3,1}^{\mathbb{R}} = 2(3-1)(1)+3 = 7$ real 3-way products.

General degree $k = \deg P$ ($d \geq 2$). Using the q-th Hasse derivative $D^{[q]}P(s_0) = \frac{1}{q!}P^{(q)}(s_0)$—simply the q-th Taylor coefficient of P around s_0, defined algebraically without division by $P(s_0+t) = \sum_{q=0}^k (D^{[q]}P)(s_0)\, t^q$—define the *reversed Hasse polynomial* $H_P(s_0; X)$ on mode 0 and the *truncated exponential polynomials* $E_k(s_j; X)$ on modes $j = 1, \ldots, d-1$:

$$H_P(s_0;X) = \sum_{q=0}^{k} q!\,(D^{[q]}P)(s_0)X^{k-q},\ E_k(s_j;X) = \sum_{q=0}^{k} \frac{s_j^q}{q!}X^q,\ Q(X) = H_P(s_0;X)\prod_{j=1}^{d-1} E_k(s_j;X). \tag{10}$$

Binomial convolution gives $[X^q]\prod_{j=1}^{d-1} E_k(s_j;X) = (s_1+\cdots+s_{d-1})^q/q!$ for all $0 \le q \le k$, so $[X^k]Q(X) = P(s_0+\cdots+s_{d-1})$ with $\deg Q \le dk$. Three coefficient-extraction schemes then yield all three upper bounds:

1. *Complex rates over* $\mathbb{C}$*:* Put $M_{d,k}^{\mathbb{C}} = (d-1)k+1$ and $\zeta = e^{2\pi i/M_{d,k}^{\mathbb{C}}}$. Because $d \ge 2$, every monomial index $n \in \{0,\dots,dk\}$ satisfies $-M_{d,k}^{\mathbb{C}} < -k \le n-k \le (d-1)k < M_{d,k}^{\mathbb{C}}$, so $M_{d,k}^{\mathbb{C}} \mid (n-k) \iff n = k$. Discrete Fourier averaging $[X^k]Q(X) = \frac{1}{M_{d,k}^{\mathbb{C}}}\sum_{r=0}^{M_{d,k}^{\mathbb{C}}-1} \zeta^{-rk}Q(\zeta^r)$ multiplied by $\prod_{j=0}^{d-1} e^{\mu s_j}$ gives $M_{d,k}^{\mathbb{C}} = (d-1)k+1$ complex products.
2. *Real rates over* $\mathbb{R}$ *(*$\operatorname{Im}\lambda = 0$*,* $P \in \mathbb{R}[X]$*):* On modes $1,\dots,d-1$, the real polynomial $A(X) = \prod_{j=1}^{d-1} E_k(s_j;X) \in \mathbb{R}[X]$ has $\deg A \le (d-1)k < M_{d,k}^{\mathbb{C}}$. Real Lagrange interpolation of $A(X)$ at the $M_{d,k}^{\mathbb{C}}$ distinct real nodes $r = 0,1,\dots,M_{d,k}^{\mathbb{C}}-1 \in \mathbb{R}$ expresses each $[X^q]A(X)$ $(0 \le q \le k)$, and hence $P(\sum_j s_j)e^{\lambda\sum_j s_j}$, as a sum of $M_{d,k}^{\mathbb{C}} = (d-1)k+1$ real d-way products.
3. *Nonreal rates over* $\mathbb{R}$ *(*$\operatorname{Im}\lambda > 0$*, Proof of Propositions 2.9 and 2.10):* Put $M = M_{d,k}^{\mathbb{R}} = 2(d-1)k+d = (2k+1)d-2k$ (note $M = d$ when $k = 0$). Writing $\operatorname{Re} w = \frac{1}{2}(\bar{w}+w)$ in (8) and expanding over subsets $S \subseteq \{0,\dots,d-1\}$ of cardinality $m = |S|$ and degree tuples $q \in \{0,\dots,k\}^d$, the identity $(-1)^r = e^{i\pi rM/M}$ produces the geometric phase $\exp(2\pi i r K(S,q)/M)$ with integer frequency

$$K(S,q) = (2k+1)m + \sum_{j\notin S} q_j - \sum_{j\in S} q_j - k \in \mathbb{Z}.$$

   Because $0 \le q_j \le k$ and $d \ge 2$: if $m = 0$ $(S = \emptyset)$, $-M < -k \le K(\emptyset,q) = \sum_j q_j - k \le (d-1)k < M$, so $M \mid K \iff \sum_j q_j = k$; if $m = d$ $(S = \{0,\dots,d-1\})$, $K = M+(k-\sum_j q_j)$, so $M \mid K \iff \sum_j q_j = k$; and if $1 \le m \le d-1$,

$$1 \le (k+1)m - k \le K(S,q) \le (k+1)m + (d-1)k \le 2(d-1)k + d - 1 = M-1,$$

   so $M \nmid K(S,q)$ and $\sum_{r=0}^{M-1} e^{2\pi i r K(S,q)/M} = 0$. Applying (8) to $U_0(X) = e^{\lambda s_0}H_P(s_0;X)$ and $U_j(X) = e^{\lambda s_j}E_k(s_j;X)$ $(j \ge 1)$ yields $M_{d,k}^{\mathbb{R}} = 2(d-1)k+d$ real d-way products.

3.3 Multilinear lower bounds via Alder–Strassen degree peeling

Step 1: From set-theoretic 1-D functions to multilinear dual functionals. Let V be a finite-dimensional $\mathbb{K}$-vector space $(\mathbb{K} \in \{\mathbb{R},\mathbb{C}\})$, let $v : \mathbb{R}_{\ge 0} \to V$ be an evaluation curve whose image spans V $(\operatorname{span}_{\mathbb{K}}(v(\mathbb{R}_{\ge 0})) = V$, verified by checking $\Lambda \circ v = 0 \implies \Lambda = 0$ for all $\Lambda \in V^*)$, and let $\mathcal{M} : V^d \to \mathbb{K}$ be d-multilinear. If the Hankel kernel $(s_0,\dots,s_{d-1}) \mapsto \mathcal{M}(v(s_0),\dots,v(s_{d-1}))$ admits a rank-R decomposition (1) with arbitrary functions $f_{n,j} : \mathbb{R}_{\ge 0} \to \mathbb{K}$, choosing a basis $\{v(t_b)\}_{b=1}^{\dim V}$ of V and defining unique linear functionals $\ell_{n,j} \in V^*$ by $\ell_{n,j}(v(t_b)) = f_{n,j}(t_b)$ yields

$$\mathcal{M}(u^{(0)},\dots,u^{(d-1)}) = \sum_{n=0}^{R-1}\prod_{j=0}^{d-1} \ell_{n,j}(u^{(j)}) \qquad \text{for all } u^{(0)},\dots,u^{(d-1)} \in V, \tag{11}$$

because both sides are d-multilinear maps agreeing on every d-tuple of basis vectors.

Step 2: Geometric intuition on $d = 3, k = 1$ **(**$\|x\|^2\cos(\|x\|^2)$ **requires 7 real terms).** Before stating the general induction, consider $K_3(x) = (x_0^2+x_1^2+x_2^2)\cos(x_0^2+x_1^2+x_2^2)$, where $v(s) = (e^{is}, se^{is}) \in \mathbb{C}^2$ spans $V = \mathbb{C}\oplus\mathbb{C}\varepsilon \cong \mathbb{C}[T]/(T^2)$ $(\varepsilon^2 = 0)$ over $\mathbb{R}$ and (11) represents the real 3-multilinear multiplication map $\mathcal{M}((u_0,u_1),(v_0,v_1),(w_0,w_1)) = \operatorname{Re}(u_1v_0w_0+u_0v_1w_0+u_0v_0w_1)$. Why must $R \ge 7$, even though difference quotients of two complex Gaussians have rank 6 for every $h \ne 0$?

- On the nilpotent subspace $E = 0 \oplus \mathbb{C}\varepsilon \cong \mathbb{R}^2$, $\mathcal{M}((0,z_1),(v_0,0),(w_0,0)) = \mathrm{Re}(z_1 v_0 w_0)$ vanishes only if $z_1 = 0$, so the restrictions $\{\ell_{n,0}|_E : 0 \le n < R\}$ must span E^* ($\dim_{\mathbb{R}} E^* = 2$). Picking two functionals (say $n = 0, 1$) forming a basis of E^*, there is a unique linear map $\gamma_0 : \mathbb{C} \to \mathbb{C}$ such that $\psi_0(u_0) := (u_0, \gamma_0(u_0)) \in \ker \ell_{0,0} \cap \ker \ell_{1,0}$ for all $u_0 \in \mathbb{C}$, *killing two terms* ($n = 0, 1$).
- Crucially, because $\varepsilon^2 = 0$ in $\mathbb{C}[T]/(T^2)$, substituting $\psi_0(u_0)$ on mode 0 and testing $(0, v_1) \in E$ on mode 1 gives $\mathcal{M}(\psi_0(u_0),(0,v_1),(w_0,0)) = \mathrm{Re}(u_0 v_1 w_0)$—*completely unaffected* by $\gamma_0(u_0)$! Thus the remaining functionals $\{\ell_{n,1}|_E : n \ge 2\}$ on mode 1 must still span E^*, letting a second map $\psi_1(v_0) = (v_0, \gamma_1(v_0))$ kill *two more terms* ($n = 2, 3$).
- Finally, evaluating $(u_0, v_0, w_1) \mapsto \mathcal{M}(\psi_0(u_0), \psi_1(v_0), (0, w_1)) = \mathrm{Re}(u_0 v_0 w_1)$ uses only the surviving $R - 4$ terms ($n \ge 4$). Since every $\mathbb{R}$-linear functional $\Lambda : \mathbb{C} \to \mathbb{R}$ has a nonzero kernel vector ($\dim_{\mathbb{R}} \mathbb{C} = 2 > 1$), induction on d shows $\mathrm{Re}(u_0 v_0 w_1)$ requires at least 3 real terms, forcing $R - 4 \ge 3$, i.e. $R \ge 7$.

Step 3: General multi-mode degree peeling over $\mathbb{C}$ and over $\mathbb{R}$. Over $\mathbb{C}$, let $V = \prod_{i\in\iota} \mathbb{C}^{\tau_i^0}$ with $\tau_i^0 = \deg P_i + 1$ and $v(s)_{i,q} = \frac{s^q}{q!} e^{\mu_i s}$. Exponential-polynomial uniqueness implies $\mathrm{span}_{\mathbb{C}}(v(\mathbb{R}_{\ge 0})) = V$, mapping the Hankel kernel to the truncated-convolution form with cutoffs $\tau_i \le \tau_i^0$:

$$\mathcal{T}_\tau(u^{(0)}, \dots, u^{(d-1)}) = \sum_{i\in\iota} \sum_{q_0+\cdots+q_{d-1}<\tau_i} w_i(q_0 + \cdots + q_{d-1}) \prod_{j=0}^{d-1} u_{i,q_j}^{(j)}, \qquad w_i(\tau_i - 1) \ne 0 \ (\tau_i > 0).$$

Strong induction on $\sum_i \tau_i$ peels one block i_0 at a time:

- When $\tau_{i_0} = 1$, testing $(e_{i_0,0}, \dots, e_{i_0,0})$ finds $\ell_{n_0,0}(e_{i_0,0}) \ne 0$; projecting block i_0 to 0 on modes $j \ge 1$ and subtracting $\frac{\ell_{n_0,0}(\pi_{\ne i_0}(x))}{\ell_{n_0,0}(e_{i_0,0})} e_{i_0,0}$ on mode 0 kills 1 term and reduces τ_{i_0} to 0.
- When $\tau_{i_0} \ge 2$, since $2(\tau_{i_0} - 1) \ge \tau_{i_0}$, rank-1 perturbations along $e_{i_0,\tau_{i_0}-1}$ on modes $j = 0, \dots, d-2$ do not cross-interact when another coordinate equals $e_{i_0,\tau_{i_0}-1}$, killing $d - 1$ distinct terms; truncating modes $0, \dots, d-2$ to $q < \tau_{i_0} - 1$ and shifting mode $d-1$ up by $+1$ ($e_{i_0,q} \mapsto e_{i_0,q+1}$) reduces $\tau_{i_0} \mapsto \tau_{i_0} - 1$ with shifted top weight $w'_{i_0}(\tau_{i_0} - 2) = w_{i_0}(\tau_{i_0} - 1) \ne 0$.

Over $\mathbb{R}$, the same induction on $G = (\prod_{i\in\iota_{\mathbb{R}}} \mathbb{R}^{\tau^0_{\mathbb{R},i}}) \times (\prod_{u\in\iota_{\mathbb{C}}} \mathbb{C}^{\tau^0_{\mathbb{C},u}})$ peels real blocks identically (1 term at $\tau_{\mathbb{R},i_0} = 1$, $d - 1$ terms at $\tau_{\mathbb{R},i_0} \ge 2$), peels nonreal $\tau_{\mathbb{C},u_0} = 1$ blocks by d terms (using $\dim_{\mathbb{R}} \ker(\Lambda : \mathbb{C} \to \mathbb{R}) \ge 1$ to kill 2 terms on mode 0 and 1 term on each mode $1, \dots, d-2$), and peels nonreal $\tau_{\mathbb{C},u_0} \ge 2$ blocks by $2(d-1)$ terms (2 independent real functionals on $0 \oplus \mathbb{C}e_{u_0,\tau_{\mathbb{C},u_0}-1} \cong \mathbb{R}^2$ per mode $j = 0, \dots, d-2$, followed by $+1$ index shifting on mode $d-1$), completing Theorem 2.5.

3.4 Proof of the general multi-Gaussian border-rank bound (Theorem 2.8)

Identity (6) is immediate from (5): for $d \ge 2$, $(k+1) + (d-2)k = (d-1)k + 1$ and $d(k+1) + (d-2)k = 2(d-1)k + d$. For a single well-formed component (λ, P) of degree $k = \deg P$ and any $h \in \mathbb{R} \setminus \{0\}$, replace $s = \sum_{j=0}^{d-1} x_j^2$ inside $P(s)$ by the exponential difference quotient $u_h(s) = \frac{e^{hs}-1}{h}$ (setting $u_0(s) := 0$ via $0^{-1} = 0$ in Lean):

$$F_h^{(\lambda,P)}(x) := \mathrm{Re}\Big(P\big(u_h(\|x\|^2)\big)\, e^{\lambda\|x\|^2}\Big).$$

Because $u_h(s) = L_h(e^{hs})$ is a degree-1 polynomial $L_h(Y) = -h^{-1} + h^{-1}Y$ in $Y = e^{hs}$, the composed polynomial $Q_h(Y) = (P \circ L_h)(Y)$ still has degree at most k:

$$P(u_h(s))\, e^{\lambda s} = \sum_{m=0}^{k} ([Y^m]Q_h)\, (e^{hs})^m e^{\lambda s} = \sum_{m=0}^{k} ([Y^m]Q_h) \prod_{j=0}^{d-1} e^{(\lambda+mh)x_j^2}.$$

When $\operatorname{Im}\lambda = 0$ (so $P, Q_h \in \mathbb{R}[X]$), this is a sum of $k+1 = b_d(\lambda, P)$ real Gaussian products; when $\operatorname{Im}\lambda > 0$, it is a sum of $k+1$ complex Gaussian products, whose real part has real d-way rank at most $d(k+1) = b_d(\lambda, P)$ by Proposition 2.9. (At $k = 1$ and $\lambda = 0$, $u_h(s) = h^{-1}(\prod_{j=0}^{d-1} e^{hx_j^2} - 1)$ realizes the classical W-tensor degeneration $[X^1]\prod_{j=0}^{d-1}(1 + s_j X)$ of exact rank d and border rank 2 [8, 14, 80] directly by smooth radial Gaussians.) Summing over $(\lambda_u, P_u) \in \mathcal{R}$ gives $\operatorname{srank}_{\mathbb{R}}(F_h) \le B_d(\mathcal{R})$ for all $h \in \mathbb{R}$. Finally, for any compact $C \subset \mathbb{R}^d$, let $B = \sup_{x\in C}\|x\|^2 < \infty$. Twice applying the mean-value inequality to $y \mapsto e^y - 1 - y$ gives $|e^y - 1 - y| \le y^2 e^{|y|}$, so $|u_h(s) - s| \le |h|(B+1)^2 e^{B+1} \to 0$ uniformly for $s \in [0, B]$ as $h \to 0$ ($h \ne 0$). Uniform continuity of $(u, s) \mapsto \operatorname{Re}(P(u)e^{\lambda s})$ on $[-1, B+1] \times [0, B]$ then implies $F_h \to \operatorname{profile}(\mathcal{R}, \|\cdot\|^2)$ uniformly on C.

4 Standard vs. Nonstandard Proofs of the Semigroup Step

Every proof of necessity above rests on the continuous one-parameter semigroup theorem [54, 22]: if $\mathcal{A}$ is a Banach algebra over $\mathbb{R}$ and $M : \mathbb{R} \to \mathcal{A}$ satisfies $M(0) = 1$, $M(s+t) = M(s)M(t)$ for all $s, t \ge 0$, and M is continuous on $[0, \infty)$, then there exists $A \in \mathcal{A}$ such that $M(t) = \exp(tA)$ for all $t \ge 0$.

4.1 Integral smoothing vs. Euler's infinitesimal step with homogeneous scale

Integral smoothing (`Semigroup.lean`). Following Nagumo [54] (see Engel and Nagel [22], Thm. I.3.7), since $M(0) = 1$ and M is continuous at 0, there exists $\delta > 0$ such that $I_\delta = \int_0^\delta M(u)\,du$ satisfies $\|I_\delta - \delta \cdot 1\| < \delta$, making I_δ invertible in $\mathcal{A}$ by Neumann series. For $t \ge 0$, changing variables in $M(t)I_\delta = \int_t^{t+\delta} M(u)\,du$ shows by the fundamental theorem of calculus that M is right-differentiable on $[0, \infty)$ with $M'(t) = M(t)A$, where $A = (M(\delta) - 1)I_\delta^{-1}$ commutes with $M(t)$. Differentiating $t \mapsto M(t)\exp(-tA)$ gives 0, so $M(t) = \exp(tA)$ in *any* Banach algebra.

Finite differences (`GeneratorBound.lean`, `FiniteDifferenceSemigroup.lean`). Without integration, if $\|M(u) - 1\| \le \frac{1}{2}$ on $[0, \delta]$, geometric-series telescoping bounds the difference quotients $\|(M(\delta/n) - 1)/(\delta/n)\|$. In finite dimensions, compactness extracts a convergent subsequence $(M(h_n) - 1)/h_n \to A$; along h_n, the one-sided difference quotients of $M(t)\exp(-tA)$ tend to 0, and a one-sided mean-value lemma shows $M(t)\exp(-tA) = 1$.

Nonstandard analysis with homogeneous common scale. Euler's heuristic sets $A = \operatorname{st}((M(\varepsilon) - 1)/\varepsilon)$ for a positive infinitesimal $\varepsilon > 0$ and iterates $M(\varepsilon)^{\lfloor t/\varepsilon \rfloor} \approx (1 + \varepsilon A)^{\lfloor t/\varepsilon \rfloor} \approx \exp(tA)$ (compare nonstandard treatments of Lie's theorems and Hilbert's fifth problem [33, 28]). Directly applying st to $(M(\varepsilon) - 1)/\varepsilon$, however, requires proving beforehand that this quotient is finite (the 124-line geometric-sum bound above).

All three nonstandard formalizations in this project eliminate that *a priori* bound by passing to *projective / homogeneous coordinates*: put $x = M(\varepsilon) - 1$ (infinitesimal by continuity at 0) and define the common infinitesimal scale

$$\eta = \varepsilon + \|x\| > 0. \tag{12}$$

Because $\frac{\varepsilon}{\eta} + \left\|\frac{x}{\eta}\right\| = 1$ with $\frac{\varepsilon}{\eta} \in [0, 1]$ and $\left\|\frac{x}{\eta}\right\| \le 1$, both rescaled quantities are bounded by 1 *without any analytical estimate*. In a finite-dimensional algebra $\mathcal{A}$, Heine–Borel compactness gives a standard vector $B = \operatorname{st}(x/\eta)$ with $\|B\| \le 1$ and $\varepsilon/\eta \approx a := 1 - \|B\| \in [0, 1]$. Comparing $N_{\text{step}} = \lfloor s/\eta \rfloor$ steps of $M(\varepsilon)$ (arriving at time $N_{\text{step}}\varepsilon \approx sa$) against N_{step} steps of $\exp(\eta B)$

Table 4: Verified source sizes for the five complete proofs of the continuous one-parameter semigroup step $M(t) = \exp(tA)$ (single-theorem case study; every project helper lemma in each proof's module dependencies is counted in full, excluding standard Mathlib and Isabelle/HOL library imports).

Proof method	Prover / files	Phys.	Nonbl.	Scope
Integral smoothing (main)	Lean 4 (`Semigroup.lean`)	96	87	Any Banach algebra
Finite differences	Lean 4 (`GeneratorBound`, `FD`)	256	233	Finite-dim.
NSA (ultrafilter seq.)	Lean 4 (`NonstandardSemigroup`)	206	190	Finite-dim.
NSA (ultrapower API)	Lean 4 (`Ultrapower*` + shared)	399	344	Finite-dim.
NSA (star + `transfer`)	Isabelle (`Semigroup_NSA.thy`)	275	253	Finite-dim. ($\|1\| = 1$)

(arriving at time $N_{\text{step}}\eta \approx s$) for an arbitrary standard $s \geq 0$,

$$\frac{\|M(\varepsilon) - \exp(\eta B)\|}{\eta} \leq \left\| \frac{x}{\eta} - B \right\| + \left\| B - \frac{\exp(\eta B) - 1}{\eta} \right\| \approx 0,$$

so summing N_{step} single-step errors of size $\eta \cdot o(1)$ with $N_{\text{step}}\eta \leq s$ yields $M(sa) = \exp(sB)$ for all standard $s \geq 0$. Could the time coordinate vanish ($a = 0$)? If $a = 0$, then $\exp(sB) = M(0) = 1$ for every $s \geq 0$; differentiating at $s = 0$ gives $B = 0$, contradicting $a + \|B\| = 1$. Hence $a > 0$, and setting $A = B/a$ with $s = t/a$ gives $M(t) = \exp(tA)$.

4.2 Three nonstandard formalizations and observational comparison

We formalized the homogeneous-scale nonstandard proof in three settings:

1. **Isabelle/HOL (`Isabelle/Semigroup_NSA.thy`):** Built on Fleuriot's nonstandard analysis development [24], Hölzl, Immler, and Huffman's typeclass hierarchy [35] (see also Unruh and Rodríguez Caballero [72] for recent operator theory in Isabelle/HOL), and Huffman's nonstandard `transfer` tactic (`transfer_principle.ML` in `HOL-Nonstandard_Analysis`). Single-step error accumulation reduces to a purely algebraic noncommutative power estimate (`power_difference_bound`): if $\|u\|, \|v\| \leq e^h$ with $h \geq 0$, then $\|u^n - v^n\| \leq ne^{nh}\|u - v\|$. Since $\|M(\varepsilon)\| \leq 1 + \|x\| \leq 1 + \eta \leq e^\eta$ and $\|\exp(\eta B)\| \leq e^{\eta\|B\|} \leq e^\eta$, `transfer` applies this ordinary inequality directly to a hypernatural step count. (This uses $\|1\| = 1$ from Isabelle's `real_normed_algebra_1`; in Lean, $\mathcal{L}(\mathcal{W}(\psi))$ has $\|1\| = 0$ when $\psi = 0$, so the Lean theorems avoid assuming $\|1\| = 1$ and telescope steps via compact-interval bounds instead.)

2. **Lean 4 with ultrafilter sequences (`Experiments/NonstandardSemigroup.lean`):** Expresses "infinitely close" directly as convergence along hyperfilter $\mathbb{N}$ (since Mathlib defines Hyperreal = Germ(hyperfilter $\mathbb{N}$) $\mathbb{R}$ without a transfer tactic), obtaining B in one line from compactness of the closed unit ball along ultrafilters.

3. **Lean 4 in nonstandard vocabulary (`Experiments/Ultrapower*.lean`):** Builds a reusable 195-line ultrapower interface (`Hyp`, `std`, `ext`, $\approx_{\text{h}}$, one-variable transfer lemmas) and writes Euler's proof in classical nonstandard notation.

Table 4 should be read as an *observational case study on a single theorem* rather than a general benchmark across proof assistants, for three reasons. First, line counts measure *library amortization* at a specific snapshot (`Mathlib v4.36.0-rc2` vs. `Isabelle2026`), not intrinsic mathematical complexity: `Semigroup.lean` (96 lines) is shortest and holds in every Banach algebra because Mathlib [50, 21] already packages Bochner integration, Neumann series, and the fundamental theorem of calculus into one-line lemmas, whereas compactness-based proofs are inherently finite-dimensional. Second, single-author comparisons carry *author-fluency and style*

Table 5: Module architecture and verified source line counts of the `KernelRankLean` library.

Module(s)	Mathematical content	Phys.	Nonbl.
`Defs`, `KernelRankLean`	Multi-way rank, surjections, slices, exact & border costs	253	215
`RowSpan`, `Complexification`	2-way row-span rank; $\mathbb{R} \to \mathbb{C}$ invariance	221	189
`Translations`	d-D radial-to-Hankel reduction; `translateSpan`	92	80
`ExpPoly`, `RealExpPoly`	Exponential-polynomial uniqueness	198	170
`Semigroup`, `TranslationSemigroup`	Banach semigroups; half-line shift semigroup	262	219
`Necessity`, `Representation`	Generalized eigenvectors; canonical conjugate pairing	222	198
`Differential`, `ComplexRank`	Half-line operators $q(D)$; 2-D complex rank	318	284
`Annihilator`	Minimal constant-coefficient differential annihilator	69	62
`MultiComplexRank`	Exact d-way complex rank $\sum_\mu((d-1)\deg P_\mu + 1)$	1,015	978
`MultiRealRank`	Exact d-way real rank (`multiTotalCost`)	2,562	2,521
`Main`	Unified d-D/2-D theorems & general border-rank bound	538	516
Total (`KernelRankLean`)		**5,750**	**5,432**

bias: standard Lean/Mathlib analysis is far better represented in AI pre-training corpora than Isabelle's `HOL-Nonstandard_Analysis`, so a specialist in Isabelle's NSA tactics might shorten `Semigroup_NSA.thy` further. Third, higher library leverage trades off against *maintenance coupling*: `Semigroup.lean` depends on Mathlib's rolling Bochner-integral and Fréchet-derivative APIs, whereas the finite-difference and Isabelle NSA proofs rely on elementary normed-algebra lemmas and Isabelle's stable annual release. Subject to those caveats, within our Lean development itself the $\approx 2\times$ gap between integral smoothing (96 lines) and ultrafilter sequences (206 lines) persisted even after eliminating the 124-line difference-quotient bound via homogeneous coordinates (12), and discounting the 195-line interface `Ultrapower.lean` leaves $169+35 = 204$ lines (confirming that without automated Łoś transfer, nonstandard vocabulary in Lean wraps filter limits). In Isabelle/HOL, `transfer` genuinely compresses hypernatural induction, while 59 lines of local helpers (vector standard parts and half-line continuity via $M(\max(t,0))$) and dual standard/nonstandard type statements bring the total to 275 lines.

5 Formalization Architecture and Verification Audit

Table 5 breaks down the `KernelRankLean` library in Lean 4 [52, 50] (5,750 physical lines, 5,432 nonblank lines; earlier commits comprised 5,578 lines at `1094bf1`, 5,324 lines at `9722ff5`, 3,822 lines at `6aabb4d` for d-D complex and multi-Gaussian ranks, 1,587 lines at `cae13d6` for 2-D plus differential annihilators, and 1,340 lines at `240dea0` for 2-D classification alone). Whereas prior formalized tensor-rank developments establish deep-network CP-rank bounds via generalized matricization in Isabelle/HOL [4] or finite-dimensional octonion and $\mathbb{F}_2$ matrix-multiplication rank bounds in Lean 4 [37, 77], our library couples continuous-domain Banach-algebra analysis with exact arbitrary-dimension real and complex multilinear rank formulas. Three engineering choices proved decisive:

1. **Dependent multi-way definitions with homogeneous two-coordinate specialization:** Defining `IsSumOfProducts` and `separableRank` over a dependent family $\prod_{i\in\iota} X_i$ unifies the 2-D and d-D theorems, while specializing `rowSpan` in `RowSpan.lean` to homogeneous $\mathrm{Fin}\,2 \to \alpha$ (via a pattern-matched helper `fin2Vec`) avoids higher-order unification timeouts on $\mathrm{Fin}\,2 \to \mathbb{R}_{\geq 0}$.

2. **Explicit uniformity selection on `translateSpan`:** Transferring a norm from $\mathbb{C}^n$ onto $\mathcal{W}(\psi) \leq (\mathbb{R}_{\geq 0} \to \mathbb{C})$ creates a second `UniformSpace` instance alongside the subtype uniformity; explicitly installing the norm-induced uniformity as the local instance prevents typeclass diamond timeouts while letting finite-dimensional linear maps inherit automatic continuity.

3. **Cayley–Hamilton divisibility in place of cyclic vectors:** Bounding $\dim_{\mathbb{C}} \mathcal{W}(\psi)$ from below via divisibility of $\chi_A(X)$ by $\prod_\mu (X-\mu)^{\deg P_\mu + 1}$ eliminated coefficient-synthesis operators and cyclic-vector constructions, shrinking the 2-D proof plus differential-annihilator extension by 93 lines while yielding Theorem 2.6 directly.

Comparison with formalized tensor-rank developments and reusable components. Recent formalizations of tensor rank address complementary settings: Bentkamp, Blanchette, and Klakow [4] formalize finite-dimensional CP-rank and generalized matricization bounds over $\mathbb{R}$ in Isabelle/HOL to separate shallow and deep convolutional arithmetic circuits; Jain [37] formalizes slice-peeling lower bounds (≥ 18, via a two-slice real pencil bound analogous to the $d = 3, k = 0$ base case of `cplxTruncForm_rank_ge`) and the Krawczyk contraction theorem in Lean 4 (checking the concrete rational certificate inequalities for rank ≤ 25 via external scripts) on the $8 \times 8 \times 8$ real octonion multiplication tensor; and Yang et al. [77] verify $R_{\mathbb{F}_2} \geq 21$ for 3×3 matrix multiplication over $\mathbb{F}_2$ via finite quotient-rank certificates in Lean 4. By contrast, `KernelRankLean` formalizes *functional* CP rank on infinite-dimensional domains $\mathbb{R}^d$ and proves exact parametric formulas for all dimensions $d \geq 2$ and degrees $k \geq 0$. Roughly two-thirds of the codebase ($\approx$ 3,900 of 5,750 lines) is general-purpose infrastructure reusable outside radial kernel theory:

- *Reusable general-purpose modules:* `Defs`, `RowSpan`, and `Complexification` (coordinate-wise `separableRank` on arbitrary finite Pi types over commutative semirings/fields, surjection invariance, and 2-way row-span rank); `Semigroup` ($M(t) = \exp(tA)$ in any real Banach algebra) and `Experiments/Ultrapower` (generic ultrapower interface); and the algebraic cores of `MultiComplexRank` and `MultiRealRank` (Propositions 2.9–2.10, the dual-span function-to-multilinear bridge `exists_linear_factors_of_span_eq_top`, and Alder–Strassen truncated-convolution / local-algebra rank lower bounds over $\mathbb{C}$ and $\mathbb{R}$).
- *Problem-specific radial-kernel glue:* `Translations`, `TranslationSemigroup`, `ExpPoly`, `RealExpPoly`, `Necessity`, `Representation`, `Differential`, `Annihilator`, and `Main` (the $s_j = x_j^2$ reduction, half-line shift semigroup on `translateSpan`, canonical upper-half-plane conjugate pairing, and multi-Gaussian difference-quotient limits).

Constructive witnesses, `noncomputable` definitions, and axioms. In Lean 4, `separableRank` is marked `noncomputable` because taking the infimum in Definition 2.1 over all univariate factor families $X_i \to \mathbb{K}$ has no general decision procedure, and Mathlib's real and complex numbers are classical quotient types. Nevertheless, all upper-bound and border-rank proofs in `MultiComplexRank.lean`, `MultiRealRank.lean`, and `Main.lean` construct *explicit closed-form* univariate factor formulas (Propositions 2.9–2.10 and Section 3.4), evaluated directly in `Applications/acoustics_3d.py`. Across all 22 audited top-level theorems, `#print axioms` reports only Lean's three standard classical axioms (`propext`, `Classical.choice`, `Quot.sound`), entering through Mathlib's Cauchy/quotient reals, coordinate surjection right inverses (2), finite-dimensional basis extraction in $\mathcal{W}(\psi)$ and V, and ultrafilter existence in `Experiments/`.

Automated verification audit. A single script (`scripts/check`, wired as the repository's pre-commit hook) builds every module of `KernelRankLean` and `Experiments` via `lake build`, verifies zero `sorry`s in compiler output, runs `#print axioms` on all 22 top-level Lean 4 theorems, builds the Isabelle2026 session `KernelRank_NSA`, and audits `Semigroup_NSA.semigroup_eq_exp` via `Thm_Deps.all_oracles` (catching `skip_proof`/`sorry` even under `quick_and_dirty`) while rejecting any `axiomatization` or `axioms` declarations.

6 Illustrative Analytical and GPU Case Studies

6.1 Experimental scope, measurement protocol, and Fourier rank invariance

Sections 6.3–6.4 are designed as *reproducible, illustrative case studies* of how Theorems 2.5–2.8 and Propositions 2.9–2.10 manifest in numerical filter design, finite-precision conditioning, and stencil cascades—not as a cross-architecture hardware benchmark suite. All scripts, GPU kernels (`Applications/gpu_kernels.py`), and JSON archives live in `Applications/` (explicit coefficients in Appendix A) and were run on one workstation: Linux `7.0.0-43-generic` (`x86_64`), Intel Core Ultra 5 125H CPU (14 cores / 18 threads, 18 MB L3 cache, 61 GiB DDR5), and NVIDIA GeForce RTX 4070 Ti Super GPU (16,376 MiB GDDR6X, 48 MB L2 cache, 672 GB/s peak bandwidth, driver `615.71.09`), with Python `3.14.4`, NumPy `2.5.4`, SciPy `1.18.1` (HiGHS LP), PyTorch `2.15.0` (CUDA `13.2`), and Triton `3.9.0`.

To separate deterministic properties from empirical variability: (i) Chebyshev pass/stopband ripples $\pm\delta$, Brillouin-octant phase-velocity errors $|\delta c/c_0|$, and 3-D spherical-pulse relative L^2 errors on the analytical D'Alembert initial condition are *deterministic* functions of the fixed coefficients in Appendix A; (ii) IEEE `float16` bokeh errors are reported both on a fixed reference scene (`seed=7`) in Table 6 and as mean $\pm$ std across 20 independent random HDR scenes (`seed=0..19`) plus HDR peak sweeps ($8\times, 48\times, 256\times$) in the four-stage ablation of Table 8; and (iii) every GPU kernel in `gpu_kernels.py` (separable, dense direct, and `cuFFT`) auto-tunes its window, block, strip/slab, and 7-smooth FFT-period parameters at startup (`config_*`), asserts numerical agreement against a `float64` reference in the same run (`err_max_*`, `err_rms_*`: max `float32` error $\le 6.1\times10^{-6}$ in 2-D and $\le 6.6\times10^{-6}$ in 3-D), and reports mean $\pm$ std across CUDA-Event batches alongside multi-core CPU reference timings (4–5 runs; the four 1-pass FDTD baselines are validated via the analytical spherical-pulse test). Between complete GPU benchmark runs, autotuner choices and buffer placement shift mean timings by up to $\approx 6\%$ in 2-D and $\approx 3\%$ in 3-D (exceeding within-run batch standard deviations), so few-percent differences between methods should not be read as strict rankings.

Before turning to discrete designs, note a continuous-domain Fourier duality: for $\operatorname{Re}\lambda < 0$, differentiating $\mathcal{F}_{\mathbb{R}^d}[e^{\lambda\|x\|^2}](\xi) = (\pi/(-\lambda))^{d/2}\exp(\|\xi\|^2/(4\lambda))$ k times with respect to λ maps $P(\|x\|^2)e^{\lambda\|x\|^2}$ $(\deg P = k)$ to $\widetilde{P}(\|\xi\|^2)e^{\|\xi\|^2/(4\lambda)}$ with $\deg\widetilde{P} = k$ and $\operatorname{Re}(1/(4\lambda)) < 0$. By Theorems 2.4 and 2.5, the d-dimensional Fourier transform preserves exact complex and real d-way separable rank across all integrable radial exponential-polynomial kernels.

6.2 Harmonic "beach-ball" differential operators and steerable kernels

Complex partial-derivative operators $(\partial_x + i\partial_y)^m$ provide a canonical rotation-equivariant basis for 2-D symmetry and steerable filters [26, 40, 7, 62]. Specializing Proposition 2.9 to equal symbolic factors $z_0 = \cdots = z_{m-1} = \partial_x + i\partial_y$ (writing $m \ge 1$ for the harmonic order and $D_\theta = \cos\theta\,\partial_x + \sin\theta\,\partial_y$ for the directional derivative along angle θ) gives, for *every* integer $m \ge 1$ (both odd and even),

$$\operatorname{Re}\big((\partial_x + i\partial_y)^m\big) = \frac{2^{m-1}}{m}\sum_{r=0}^{m-1}(-1)^r D^m_{\pi r/m}. \tag{13}$$

On a regular $2m$-point ring $\theta_n = \pi n/m$ $(n = 0,\dots,2m-1)$ of radius $h \to 0$, antipodal points satisfy $D_{\theta_{r+m}} = -D_{\theta_r}$ while Fourier orthogonality of $(-1)^r = e^{i\pi r}$ annihilates all Taylor orders below m. For $f \in C^m$ near (x, y), (13) recovers—by elementary geometric series rather than Cauchy residues and Chebyshev partial fractions—both the odd-order $(m = 2N+1)$ hexagonal/decagonal "beach-ball" operators of Niemitalo [58] and the even-order operators of

Perona [61] in a single formula:

$$\lim_{h\to 0}\frac{1}{h^m}\sum_{n=0}^{2m-1}(-1)^n f\Big(x+h\cos\frac{\pi n}{m},\, y+h\sin\frac{\pi n}{m}\Big)=\frac{1}{(m-1)!\,2^{m-2}}\,\mathrm{Re}\big((\partial_x+i\partial_y)^m\big)f(x,y),$$

with a half-sector ring rotation $+\frac{\pi}{2m}$ yielding $+\mathrm{Im}((\partial_x+i\partial_y)^m)$. Recognizing $\partial_{\bar z}=\frac{1}{2}(\partial_x+i\partial_y)$ as the Cauchy–Riemann operator also explains why applying $\partial_{\bar z}^m$ to *any* smooth radial kernel $\varphi(x^2+y^2)=\varphi(z\bar z)$ produces no lower-order radial derivatives [7, 62]: since $\partial_{\bar z}(z)=0$, induction gives

$$\left(\frac{\partial_x+i\partial_y}{2}\right)^m\varphi(x^2+y^2)=(x+iy)^m\,\varphi^{(m)}(x^2+y^2)=r^m e^{im\theta}\,\varphi^{(m)}(r^2). \tag{14}$$

When $\varphi(s)$ is a well-formed multi-Gaussian profile with n_{re} real and n_{cx} nonreal rates (all $\lambda_u\neq 0$, approximating, e.g., an ideal circular low-pass filter [59]), $\varphi^{(m)}(s)$ is *again* a well-formed multi-Gaussian profile at the same rates (weighted by λ_u^m) of continuous 2-way rank $R=n_{\mathrm{re}}+2n_{\mathrm{cx}}$. Analytically differentiating the continuous-space kernel on $\mathbb{R}^2$—avoiding discrete finite-difference distortion up to Nyquist [57]—absorbs the monomials of $\mathrm{Re}((x+iy)^m)$ and $\mathrm{Im}((x+iy)^m)$ into the 1-D Gaussian factors, giving exact *continuous-domain* 2-way real separable ranks on $\mathbb{R}^2$:

$$\big(\lfloor m/2\rfloor+1\big)(n_{\mathrm{re}}+2n_{\mathrm{cx}})\quad\text{for }\cos(m\theta),\qquad \lceil m/2\rceil(n_{\mathrm{re}}+2n_{\mathrm{cx}})\quad\text{for }\sin(m\theta).$$

Sampling those 1-D factors on a finite grid yields a discrete separable realization with at most that many terms. On the 161×161 ($\omega_c=\pi/64$) orientation-estimation filter of [57], replacing direct $2\times 161^2=51{,}842$ taps/pixel by a two-complex-Gaussian ($R=4$) analytical gradient pair reduces scalar multiply–adds to 2,576 (20×, or 40× with even/odd tap-parity folding).

Continuous-domain pass-count bounds for simultaneous steerable pairs. Can sharing intermediate 1-D continuous convolutions reduce simultaneous evaluation of (C_m*I, S_m*I) below separate evaluation $2(\lfloor m/2\rfloor+1)R+2\lceil m/2\rceil R=2(m+1)R$? No: if any two-stage 1-D schedule on $\mathbb{R}^2$ splits $(C_m,S_m)=(A_c+B_c, A_s+B_s)$ into P_y shared vertical passes followed by $\mathrm{rank}(A_c)+\mathrm{rank}(A_s)$ horizontal passes plus P_x shared horizontal passes followed by $\mathrm{rank}(B_c)+\mathrm{rank}(B_s)$ vertical passes, then

$$\begin{aligned}\dim\mathrm{rowSpan}_x(C_m,S_m)&\le\mathrm{rank}(A_c)+\mathrm{rank}(A_s)+P_x,\\ \dim\mathrm{colSpan}_y(C_m,S_m)&\le P_y+\mathrm{rank}(B_c)+\mathrm{rank}(B_s).\end{aligned}$$

Because C_m and S_m on $\mathbb{R}^2$ span all powers $\{x^0,\dots,x^m\}g_n(x)$ in x and $\{y^0,\dots,y^m\}g_n(y)$ in y, adding the two inequalities shows that *every* two-stage continuous 1-D schedule requires at least $2(m+1)R$ passes ($4R$ passes for (h_x,h_y)). Savings instead come from: (i) definite even/odd parity of every 1-D factor $u^q g_n(u)$, halving multiplications per sampled pass via $I[y+\ell]\pm I[y-\ell]$ folding; (ii) sharing $\{I*_y(y^q g_n)\}_{q=0}^M$ across *multiple* harmonic orders ($5R$ instead of $6R$ for $m=0,1$; $10R$ instead of $12R$ for $m=1,3$); and (iii) for the d-D gradient $(h_{x_0},\dots,h_{x_{d-1}})$ of a multi-Gaussian radial kernel ($d\ge 3$, base rank $R_d=n_{\mathrm{re}}+d\,n_{\mathrm{cx}}$), recursively splitting $\{0,\dots,d-1\}$ into halves to reduce d^2R_d passes to $T(d)R_d$ with $T(1)=1$ and $T(d)=T(\lfloor d/2\rfloor)+T(\lceil d/2\rceil)+d\le d(\lceil\log_2 d\rceil+1)$ ($8R_3$ vs. $9R_3$ in $d=3$; $12R_4$ vs. $16R_4$ in $d=4$).

6.3 Worked application I: 2-D circular lens blur and $\mathrm{GL}_R(\mathbb{R})$ conditioning

In depth-of-field bokeh rendering [56, 27, 60], the ideal defocus point-spread function is the discontinuous unit disc $\mathbf{1}_{r\le 1}$ ($r=\sqrt{x^2+y^2}$), which has infinite separable rank: for any $0<r_1<\cdots<r_M<1$, evaluating $\mathbf{1}_{x_i^2+y_j^2\le 1}$ at $x_i=r_i$, $y_j=\sqrt{1-r_j^2}$ produces an invertible $M\times M$ upper-triangular binary matrix. Introducing a transition band $[1,S]$ with $S=1+\Delta r$ ($\Delta r=0.2$)

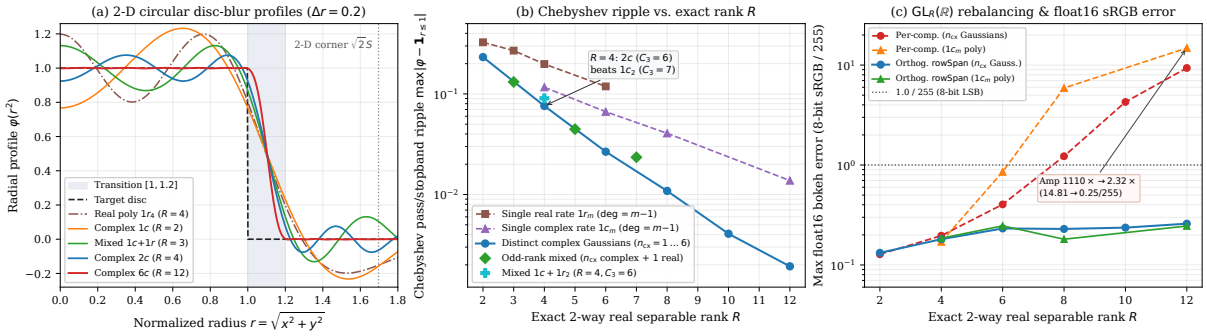


Figure 2: Two-dimensional circular disc blur ($\Delta r = 0.2$, `Applications/blur_2d.py`): (a) minimax radial profiles $\varphi(r^2)$ against the unit disc; (b) minimax pass/stopband ripple across exact 2-way real separable ranks $R \in [2, 12]$ for distinct complex Gaussians, odd-rank mixed Gaussians, and repeated-rate polynomial-Gaussians; (c) maximum IEEE `float16` bokeh error in 8-bit sRGB units (/255, via IEC 61966-2-1 sRGB encoding on a 512×512 linear-light HDR scene with night shadows 0.001 and 3×3 glints up to 48.0, $R_{\text{px}} = 25$, $W = 61$) before and after nuclear-norm-optimal $\mathrm{GL}_R(\mathbb{R})$ rebalancing on rowSpan(K).

Table 6: Representative 2-D circular disc-blur designs ($\Delta r = 0.2$, $R_{\text{px}} = 25$, $W = 61$; explicit parameters in Table 11). Exact 2-D rank $R = C_2(\mathcal{R})$ and 3-D ball rank $C_3(\mathcal{R})$ follow Theorem 2.5; SVD is the empirical singular-value count ($\sigma_j > 10^{-10}\sigma_1$) of the 61×61 kernel; $\kappa_\infty = (\sum_n \|h_n\|_\infty \|v_n\|_\infty)/\|K\|_\infty$ measures peak channel amplification before (raw) and after (orth) rowSpan rebalancing, alongside max `float16` HDR bokeh error in 8-bit sRGB units (/255, IEC 61966-2-1).

Structure	Degrees	R (d=2)	SVD	C_3 (d=3)	Ripple $\pm\delta$	$\kappa_\infty^{\text{raw}}$	$\kappa_\infty^{\text{orth}}$	sRGB raw	sRGB orth
$1c$ (Gaussian)	0	2	2	3	0.2324	1.65	1.30	0.13	0.13
$1c + 1r$ (mixed)	0, 0	3	3	4	0.1315	3.45	1.65	0.16	0.20
$2c$ (Gaussians)	0, 0	4	4	6	0.0759	5.28	1.95	0.20	0.18
$1c + 1r_2$ (mixed poly)	0, 1	4	4	6	0.0905	4.50	1.92	0.12	0.23
$1c_2$ (complex poly)	1	4	4	7	0.1165	5.68	1.88	0.17	0.19
$1r_4$ (real Laguerre)	3	4	4	7	0.1982	4.74	1.56	0.21	0.23
$2c + 1r$ (mixed)	0, 0, 0	5	5	7	0.0446	8.73	2.08	0.26	0.20
$4c$ (Gaussians)	0^4	8	8	12	0.0109	70.06	2.31	1.22	0.23
$6c$ (Gaussians)	0^6	12	12	18	0.0019	595.01	2.49	9.35	0.26
$1c_6$ (complex poly)	5	12	12	23	0.0138	1,109.75	2.32	14.81	0.25

and truncating each 1-D pass to $|x|, |y| \le S$ restricts the 2-D separable support $[-S, S]^2$ to radii $r \le \sqrt{2}\,S \approx 1.697$. For any fixed rates (α_u, ω_u), minimizing the sampled Chebyshev ripple $\max_{r \in [0,1] \cup [S, \sqrt{2}S]} |\varphi(r^2) - \mathbf{1}_{r \le 1}|$ (2,000 radial nodes) over all polynomial weights is a linear program, solved with HiGHS inside differential-evolution/Nelder–Mead rate search in `Applications/blur_2d.py`.

Figure 2 and Table 6 highlight four structural findings:

1. **Role of nonreal rates, odd budgets, and 3-D lifting:** Across all 18 tested families, empirical SVD rank matches $C_2(\mathcal{R})$ identically. Distinct complex Gaussians achieve geometric ripple decay with $R = 2n_{\text{cx}}$ (0.2324 → 0.0019 from $1c$ to $6c$, or 0.0014 under global Remez search [60]), while adding a single real Gaussian ($1c+1r, 2c+1r$) interpolates cleanly at odd ranks $R = 3, 5$. Free real-only rates coalesce into repeated-rate Laguerre–Gaussians ($1r_m$). At $R = 4$, four distinct structures tie in 2-D cost, but lifting to a 3-D spherical ball splits their exact 3-way ranks: $2c$ gives both the lowest ripple (± 0.0759) and lower 3-D rank ($C_3 = 6$ vs. $C_3 = 7$ for $1c_2$ and $1r_4$).

2. **Transitive $\mathrm{GL}_R(\mathbb{R})$ freedom and four-stage `separableChannels` ablation (Table 8):** While $R = \dim \mathrm{rowSpan}(K)$ is fixed, in $d = 2$ writing the $W \times W$ symmetric kernel matrix as $K = HV^\top$ ($H, V \in \mathbb{R}^{W \times R}$) allows replacing (H, V) by $(HT, VT^{-\top})$ for *any* invertible matrix $T \in \mathrm{GL}_R(\mathbb{R})$ [70]. To remove inter-component cancellation ($\kappa_\infty^{\mathrm{raw}} = 595$ on $6c$ and 1,110 on $1c_6$) without a full $W \times W$ SVD, compute a thin QR factorization $H = QR_H$ ($Q^\top Q = I_R$), project $K = QMQ^\top$ onto the $R \times R$ symmetric matrix $M = R_H V^\top Q$, and diagonalize $M = U \,\mathrm{diag}(e_1, \ldots, e_R) U^\top$. Setting $G = QU = [g_1, \ldots, g_R]$ yields $K(x, y) = \sum_{n=1}^{R} e_n g_n(x) g_n(y)$, where $\tilde{h}_n = \sqrt{|e_n|} g_n$ and $\tilde{v}_n = \mathrm{sign}(e_n) \tilde{h}_n$ achieve the nuclear-norm minimum $\sum_{n=1}^{R} \|\tilde{h}_n\|_2 \|\tilde{v}_n\|_2 = \|K\|_{S_1}$ ($\kappa_\infty \le 2.50$). Factoring $T \in \mathrm{GL}_R(\mathbb{R})$ into pure diagonal pair scaling, non-diagonal orthogonalization, and channel permutation in Table 8 isolates each mechanism across 20 random HDR scenes: (i) diagonal pair balancing $\|c_n h_n\|_2 = \|v_n / c_n\|_2$ (Stage 1 → Stage 2, leaving every rank-1 matrix $h_n v_n^\top$ invariant) cures `float16` intermediate overflow when degree-5 weights sit on the first pass ($1c_6$: 20/20 overflow → 15.17±1.39/255), yet *leaves cross-component cancellation intact* (7.90±0.92/255 on $6c$, 15.17 ± 1.39/255 on $1c_6$); (ii) cross-component $\mathrm{GL}_R(\mathbb{R})$ orthogonalization (Stage 2 → Stage 3) cuts 20-scene sRGB max error by 54% on $4c$ (0.54 → 0.25/255), 96.2% on $6c$ (7.90 → 0.30/255), and 98.2% on $1c_6$ (15.17 → 0.28/255) (at $R \le 4$, raw and orthogonalized errors both sit near the ≤ 0.24/255 quantization floor); and (iii) reversing accumulation from largest-$|e_n|$-first to smallest-$|e_n|$-first (Stage 3 → Stage 4) cuts *linear-light* highlight error by 1.5×–2.9× across all designs and glint peaks (0.40 → 0.15/255 at 48× and 1.41 → 0.60/255 at 256× on $6c$, and 0.23 → 0.08/255 in sRGB at unclipped 8×; at 48× and 256×, sRGB clips interior disc highlights > 1.0, so unclipped rim pixels govern sRGB max error near 0.24–0.31/255 at 48× and 1.07–1.76/255 at 256×). By contrast, in $d \ge 3$, minimal d-way CP decompositions lack transitive $\mathrm{GL}_R(\mathbb{R})$ cross-component rotations [43], making well-separated rates essential.

3. **GPU and CPU throughput vs. tuned direct W^2 stencils and global `cuFFT` (Table 9):** In `Applications/gpu_kernels.py`, all three 1920 × 1080 RGBA GPU paths are tuned around the 48 MB L2 cache: `SeparableBlur2D` evaluates all R `rowSpan`-orthogonalized terms per pass ($2c/4c/6c$ for $R = 4, 8, 12$) using K-output register sliding windows over transposed L2-resident row strips; `DirectBlur2D` is a dense $W \times W$ Triton kernel sharing loaded row segments across outputs; and `FFTBlur2D` packs RGBA channel pairs into two complex planes for complex 2-D `cuFFT` (`fft2`/`ifft2` with in-place real spectral multiply) at a tuned 7-smooth period ($\ge \mathrm{size} + L$ for zero padding, $\ge \mathrm{size} + 2L$ for clamped borders). Notably, direct $W \times W$ and `cuFFT` runtimes depend *only* on (R_{px}, W)—they evaluate any symmetric kernel on $[-L, L]^2$ (whether $2c$, $6c$, or an unapproximated disc) in the same time, whereas separable FIR cost grows with R. At $R_{\mathrm{px}} = 64$, the separable cascade beats the tuned direct $W \times W$ kernel by 28.0× (`fp32`) / 29.5× (`fp16`) at $R = 4$, 13.9× / 14.6× at $R = 8$, and 9.2× / 9.4× at $R = 12$ (whereas at $R_{\mathrm{px}} = 8$, direct 0.52 ms beats $R \ge 8$). Against `cuFFT` (0.451–0.555 ms across all radii), the separable cascade wins only at $R = 4$ for $R_{\mathrm{px}} \in \{8, 16\}$ in both precisions (0.331 / 0.417 ms vs. 0.457 ms; 0.354 / 0.396 ms vs. 0.454 ms; at $R_{\mathrm{px}} = 32, R = 4$, `fp16` 0.496 ms and clamp `cuFFT` 0.481 ms are within run-to-run variation), while `cuFFT` is both faster and free of finite-rank approximation ripple for every $R \ge 8$ and for $R = 4$ at $R_{\mathrm{px}} = 64$. Because strip intermediates stay in L2 rather than DRAM, `float16` is no longer uniformly faster than `float32` (`fp16`/`fp32` time ratios span 0.84× to 1.26× due to 16-bit gather/store overhead, though Table 8's conditioning remains essential whenever textures or APIs mandate `float16`). On the 14-core CPU (`float32`), by contrast, unfused $2R$-pass PyTorch `conv2d` writes 33 MB frame buffers to DDR5 RAM (169.1, 286.7, 444.0 ms at $R = 4$ vs. `PocketFFT` `rfft2` at 32.6, 40.7, 68.1 ms and direct `conv2d` at 68.2, 208.2, 676.6 ms for $R_{\mathrm{px}} = 8, 16, 32$). Separable decompositions at large radii are thus most compelling with $O(1)$ recursive IIR Gaussians [19, 78], spatially varying depth-of-field radii $R_{\mathrm{px}}(x, y)$ [27], or L2/line-buffered

Table 7: Sampled continuous complex Gaussians vs. direct discrete rank-R optimization across stencil radii $R_{\text{px}} \in \{3, 6, 12, 25, 64\}$ ($L + 1 = \lceil 1.2\, R_{\text{px}} \rceil + 1$ even 1-D degrees of freedom). Each entry lists pass/stopband Chebyshev ripple $\pm\delta$, with equal-radius shell anisotropy Δ_{sh} in parentheses where nonzero.

R_{px}	W	$L{+}1$	R	Cont. Gaussian	Direct ℓ_2 SVD (Δ_{sh})	Disc. unconstr. (Δ_{sh})	Disc. shell-iso.
3	9	5	4	0.0759 (0)	0.0000 (0)	0.0000 (0)	0.0000
6	17	9	4	0.0759 (0)	0.0218 (0.011)	0.0048 (0.129)	0.0048
12	31	16	4	0.0759 (0)	0.0950 (0.107)	0.0190 (0.158)	0.0427
25	61	31	4	0.0759 (0)	0.0956 (0.177)	0.0523 (0.091)	0.0737
64	155	78	4	0.0759 (0)	0.1030 (0.192)	0.0665 (0.032)	0.0758
12	31	16	8	0.0109 (0)	0.0098 (0.014)	0.0014 (0.024)	0.0008
25	61	31	8	0.0109 (0)	0.0129 (0.015)	0.0049 (0.023)	0.0055
64	155	78	8	0.0109 (0)	0.0163 (0.019)	0.0076 (0.011)	0.0107
64	155	78	12	0.0019 (0)	0.0048 (0.006)	0.0010 (0.007)	0.0017

pipelines without spectral padding buffers.

4. **Direct discrete rank-R optimization vs. sampled continuous profiles (Figure 3, Table 7):** Because any finite-tap separable cascade on $\{-L, \ldots, L\}^d$ is rectangular-truncated and even in every coordinate ($h_n[-m] = h_n[m]$), its 1-D factors lie in an $(L+1)$-dimensional even subspace, capping 2-D discrete rank at $\min(R, L+1)$ (for instance, sampling $6c$ of continuous rank 12 gives exact discrete SVD rank $4, 6, 11, 12, 12, 12$ at $R_{\text{px}} = 2, 4, 8, 16, 25, 64$, with σ_{12}/σ_1 locking onto 4.37–4.34×10^{-4} for $R_{\text{px}} \geq 16$). Does optimizing discrete rank-R matrices directly on $\{-L, \ldots, L\}^2$ improve on sampling continuous complex Gaussians? Table 7 and Figure 3 compare sampled continuous Gaussians against (i) direct Eckart–Young ℓ_2 SVD truncation of a raised-cosine smooth disc on $[1, 1.2]$, (ii) unconstrained D_4-symmetric discrete rank-R minimax optimization (alternating pass/stopband clamping with SVD rank-R projection), and (iii) shell-isotropic discrete rank-R minimax (additionally equalizing lattice points of equal Euclidean radius $m_x^2 + m_y^2 = N$, measuring shell anisotropy $\Delta_{\text{sh}} = \max_N(\max_{\|\mathbf{m}\|^2=N} K - \min_{\|\mathbf{m}\|^2=N} K)$). Three regimes emerge clearly:

   - *Small stencils ($R_{\text{px}} \leq 6$, $L + 1 \leq 9$):* When R is comparable to $L + 1$, direct discrete optimization exploits the tiny $(L+1)$-dimensional space (0.0000 error once $R \geq L + 1$, and ± 0.0048 vs. continuous ± 0.0759 at $R_{\text{px}} = 6, R = 4$; notice in $\{0, \ldots, 7\}^2$ only $1^2 + 7^2 = 5^2 + 5^2 = 50$ shares a squared radius, so even shell-isotropic discrete optimization retains 0°-vs.-45° angular freedom in Figure 3).
   - *Direct ℓ_2 SVD vs. Chebyshev minimax:* At $R_{\text{px}} \geq 12$, ℓ_2 SVD truncation degrades in both pass/stopband ripple (0.0956–0.1030 vs. 0.0759 at $R = 4$) and severe 17.7%–19.2% equal-radius shell anisotropy, producing visible cross-streaks around the 12× HDR star in Figure 3.
   - *Anisotropy trade-off and large-stencil convergence ($R_{\text{px}} \geq 25$):* Unconstrained discrete minimax lowers pass/stopband ripple (0.0523 at $R_{\text{px}} = 25, R = 4$) *only by warping the unconstrained transition rim* $1 < r < 1.2$ *anisotropically* ($\Delta_{\text{sh}} = 0.091$ at $R_{\text{px}} = 25$, 0.158 at $R_{\text{px}} = 12$; see the 0° blue vs. 45° red split and squarish $K = 0.5$ rim contour in Figure 3). Once equal-radius shell isotropy is enforced, discrete optimization converges right onto the continuous complex-Gaussian curve (0.2324 vs. 0.2324 at $R = 2$, 0.0758 vs. 0.0759 at $R = 4$, 0.0107 vs. 0.0109 at $R = 8$, and 0.0017 vs. 0.0019 at $R = 12$ at $R_{\text{px}} = 64$), without sacrificing continuous radius rescalability.

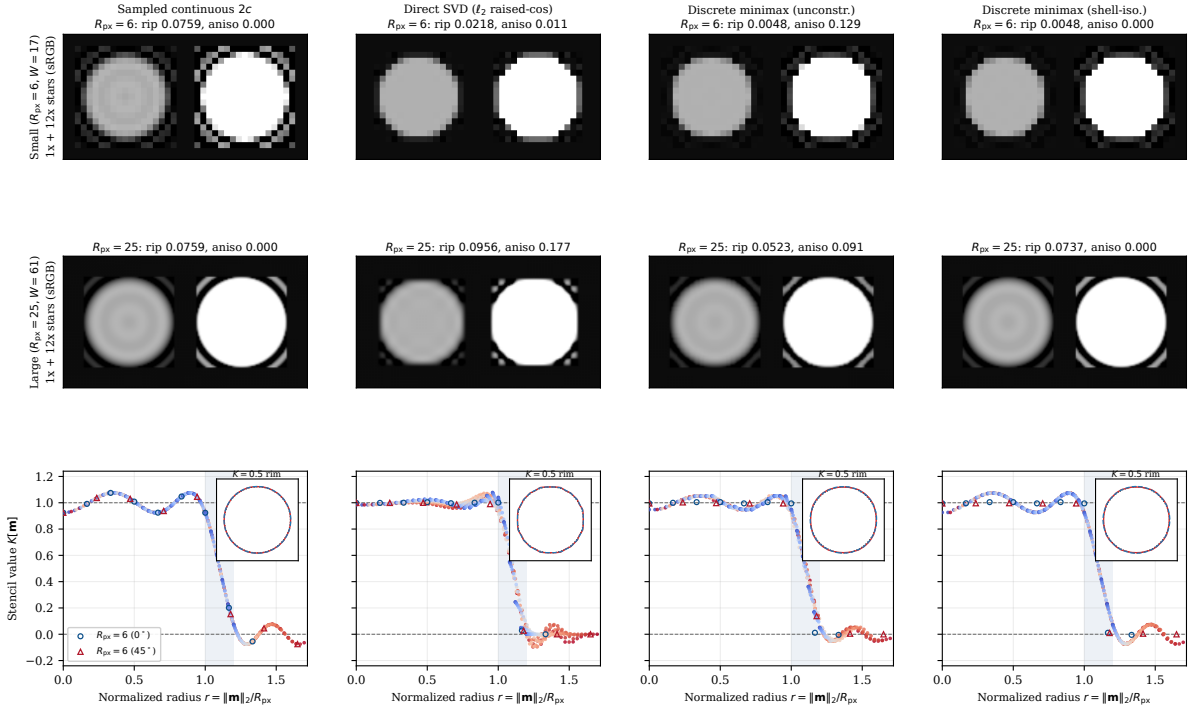


Figure 3: Visual and quantitative comparison at exact rank $R = 4$ between sampled continuous $2c$ Gaussians, direct ℓ_2 SVD of a raised-cosine disc, unconstrained discrete rank-4 minimax, and shell-isotropic discrete rank-4 minimax (`Applications/blur_2d.py`): top row, small stencil ($R_{\text{px}} = 6, W = 17$) sRGB bokeh on 1× (left) and 12× HDR (right) point stars; middle row, large stencil ($R_{\text{px}} = 25, W = 61$) sRGB bokeh; bottom row, per-pixel radial scatter ($R_{\text{px}} = 25$ dots colored from 0° blue to 45° red; $R_{\text{px}} = 6$ open 0°/45° markers) and $K = 0.5$ rim contours ($R_{\text{px}} = 6$ solid red, $R_{\text{px}} = 25$ dashed blue, reference circle dotted black).

6.4 Worked application II: isotropic 3-D acoustic wave simulation

Consider the 3-D constant-speed scalar acoustic wave equation $\partial_{tt} p = c_0^2 \nabla^2 p$ discretized on $\mathbb{Z}^3$ with spacing Δx and time step $\Delta t = \nu \Delta x / c_0$ (Courant number $\nu > 0$) by the explicit two-step leapfrog update

$$p^{n+1}[\mathbf{m}] + p^{n-1}[\mathbf{m}] = (M * p^n)[\mathbf{m}], \qquad \mathbf{m} \in \mathbb{Z}^3. \tag{15}$$

Substituting a discrete plane wave $p^n[\mathbf{m}] = e^{i(\mathbf{k}\cdot\mathbf{m} - \omega_{\text{num}} n \Delta t)}$ with normalized wavenumber $\mathbf{k} = (k_x, k_y, k_z) \in [-\pi, \pi]^3$ turns the left-hand side of (15) into $2\cos(\omega_{\text{num}}\Delta t)\, p^n[\mathbf{m}]$ and the right-hand side into $\widehat{M}(\mathbf{k})\, p^n[\mathbf{m}]$, where $\widehat{M}(\mathbf{k}) = \sum_{\mathbf{m}} M[\mathbf{m}] e^{-i\mathbf{k}\cdot\mathbf{m}}$ is the discrete Fourier symbol of M. Since an exact continuous plane wave travels at speed c_0 ($\omega_{\text{exact}}\Delta t = c_0(\|\mathbf{k}\|_2/\Delta x)\Delta t = \nu\|\mathbf{k}\|_2$), matching the exact wave equation requires $\widehat{M}_{\text{exact}}(\mathbf{k}) = 2\cos(\nu\|\mathbf{k}\|_2)$ [67], and the numerical-to-exact phase-velocity ratio is $c_{\text{num}}(\mathbf{k})/c_0 = \arccos(\widehat{M}(\mathbf{k})/2)/(\nu\|\mathbf{k}\|_2)$. Whenever $|\widehat{M}(\mathbf{k})| \le 2$, both roots of $\zeta^2 - \widehat{M}(\mathbf{k})\zeta + 1 = 0$ have $|\zeta| = 1$ (spectral von Neumann stability); strict interior containment $\widehat{M}(\mathbf{k}) \in (-2, 2)$ for $\mathbf{k} \neq \mathbf{0}$ avoids repeated roots $\zeta = \pm 1$ (linear-in-n secular growth) and yields a conserved positive discrete energy on all nonzero modes. Standard 7-point FDTD ($\nu \le 1/\sqrt{3}$) suffers 19.88% axial phase-velocity error by $\|\mathbf{k}\|_2 = 0.80\pi$, motivating dispersion-relation-preserving and high-order room-acoustics FDTD designs [68, 42, 74, 31, 32]. Two structural facts explain why radial multi-Gaussians are natural here:

- **Inadequacy of additive axial stencils of any order:** Any purely axial scheme $\widehat{M}_{\text{ax}}(\mathbf{k}) = 2 + A(k_x) + A(k_y) + A(k_z)$ ($A(0) = 0$) lacks mixed cross-terms $k_x^2 k_y^2$: even if $A(k) = 2(\cos(\nu k) -$

Table 8: Four-stage `separableChannels` IEEE `float16` conditioning ablation ($R_{\text{px}} = 25$, $W = 61$, 512×512 HDR scenes): Stage 1 puts weights on H and unit basis on V; Stage 2 rescales each channel diagonally so $\|h_n\|_2 = \|v_n\|_2$ without mixing channels; Stages 3 and 4 apply $\mathrm{GL}_R(\mathbb{R})$ orthogonalization summed largest-$|e_n|$-first vs. smallest-$|e_n|$-first. Left columns: 20-scene mean $\pm$ std max 8-bit sRGB error (/255, peak 48×); right columns: Stage 3 → Stage 4 *linear-light* max error ($255 \cdot \max|\Delta I_{\text{lin}}|$) at peaks 8×, 48×, 256× (`seed=7`).

	20-scene max 8-bit sRGB error (/255, peak 48×)				Linear max error (/255): Desc → Asc		
Design (R)	1. Unbalanced	2. Pair-bal.	3. Orth. (desc)	4. Orth. (asc)	Peak 8×	Peak 48×	Peak 256×
$2c$ ($R = 4$)	0.12 ± 0.02	0.11 ± 0.01	0.20 ± 0.04	0.24 ± 0.08	$0.24 \to \mathbf{0.13}$	$0.24 \to \mathbf{0.16}$	$1.00 \to \mathbf{0.60}$
$4c$ ($R = 8$)	0.58 ± 0.07	0.54 ± 0.06	0.25 ± 0.07	0.26 ± 0.10	$0.29 \to \mathbf{0.14}$	$0.36 \to \mathbf{0.16}$	$1.23 \to \mathbf{0.69}$
$6c$ ($R = 12$)	7.80 ± 1.16	7.90 ± 0.92	0.30 ± 0.06	0.31 ± 0.13	$0.38 \to \mathbf{0.13}$	$0.40 \to \mathbf{0.15}$	$1.41 \to \mathbf{0.60}$
$1c_6$ ($R = 12$)	`ovf` (20/20)	15.17 ± 1.39	0.28 ± 0.05	0.27 ± 0.10	$0.32 \to \mathbf{0.15}$	$0.40 \to \mathbf{0.16}$	$1.45 \to \mathbf{0.62}$

Table 9: GPU execution times in milliseconds (mean over 10 CUDA-Event batches) for 1920×1080 RGBA 2-D blur on an RTX 4070 Ti Super across $R_{\text{px}} \in \{8, 16, 32, 64\}$ ($W = 2L + 1$, $L = \lceil 1.2\, R_{\text{px}} \rceil$): tuned dense $W \times W$ Triton (`DirectBlur2D`, rank-independent cost, std $\leq$ 0.13 ms, $\leq$ 2.1%), complex-plane-packed `cuFFT` at 7-smooth zero- and clamp-padded periods (`FFTBlur2D`, rank-independent cost, std $\leq$ 0.002 ms), and two-pass register-windowed L2-strip Triton cascades on $2c/4c/6c$ (`SeparableBlur2D`, `fp32` std $\leq$ 0.013 ms, `fp16` std $\leq$ 0.032 ms; run-to-run variation $\leq$ 6%, full metadata in `Applications/results_blur_2d.json`).

R_{px}	W	Direct W^2	`cuFFT` (zero)	`cuFFT` (clamp)	Sep. $R = 4$		Sep. $R = 8$		Sep. $R = 12$
		`fp32`	`fp32`	`fp32`	`fp32`	`fp16`	`fp32`	`fp16`	`fp32` / `fp16`
8	21	0.52	0.465	0.457	**0.331**	0.417	0.665	0.670	0.860 / 1.033
16	41	1.75	0.451	0.454	**0.354**	0.396	0.719	0.682	1.076 / 1.143
32	79	6.45	**0.452**	0.481	0.528	0.496	1.121	0.937	1.661 / 1.522
64	155	24.38	**0.473**	0.555	0.871	0.825	1.755	1.672	2.641 / 2.594

1) has zero dispersion along the coordinate axes, along $(1,1,1)/\sqrt{3}$ it gives $2{+}6(\cos(\nu k/\sqrt{3}) - 1) = 2 - \nu^2 k^2 + \frac{1}{36}\nu^4 k^4 - \ldots \neq 2\cos(\nu k) = 2 - \nu^2 k^2 + \frac{1}{12}\nu^4 k^4 - \ldots$, leaving an irreducible $O(\nu^4 k^4)$ anisotropy gap while Brillouin-corner stability $\widehat{M}_{\text{ax}}(\pi, \pi, \pi) \geq -2$ forces $A(\pi) \geq -4/3$ ($\nu k_c \leq \arccos(1/3) \approx 0.392\pi$ when A tracks $2(\cos(\nu k) - 1)$).

- **Kowalczyk–van Walstijn IWB is a single real Gaussian:** In room acoustics, Kowalczyk and van Walstijn's 27-point Interpolated Wideband (`IWB`) stencil [42] ($a = \frac{1}{4}, b = \frac{1}{16}$, $\nu \leq 1$) has face, edge, and corner weights $\frac{1}{4}\nu^2, \frac{1}{8}\nu^2, \frac{1}{16}\nu^2$ and center weight $2 - \frac{7}{2}\nu^2$, which is *identically*

$$M_{\text{IWB}}[\mathbf{m}] = (2 - 4\nu^2)\delta_{\mathbf{0}}[\mathbf{m}] + \frac{\nu^2}{2}\, 2^{-(m_x^2 + m_y^2 + m_z^2)} \qquad (\mathbf{m} \in \{-1, 0, 1\}^3),$$

the ($n_{\text{re}} = 1, n_{\text{cx}} = 0, L = 1, R = 1$) single-real-Gaussian member of our radial family (plus a zero-pass center tap). Because a single positive real Gaussian cannot match higher radial derivatives of $2\cos(\nu\|\mathbf{k}\|_2)$, `IWB` at $\nu = 1$ is exact on axes but incurs 9.38% diagonal phase-velocity error by 0.80π (Figure 4a,d).

Brillouin-octant Chebyshev optimization and exact Neumann walls. Generalizing `IWB` to wider stencils $\mathbf{m} \in [-L, L]^3$, we set $M[\mathbf{m}] = \alpha_0 \delta_{\mathbf{0}}[\mathbf{m}] + \varphi(m_x^2 + m_y^2 + m_z^2)$ (parameters in Table 12), where α_0 fixes $\widehat{M}(\mathbf{0}) = 2$ identically and φ has exact 3-way real rank $R = n_{\text{re}} + 3n_{\text{cx}}$ (Proposition 2.9) or $R = 7$ for $\deg P = 1$ (Proposition 2.10; verified to $\leq 5.6 \times 10^{-16}$ on $[-L, L]^3$). For fixed rates (searched via differential evolution and Nelder–Mead), minimizing $\max_{0<\|\mathbf{k}\|_2 \leq k_c} |\widehat{M}(\mathbf{k}) - 2\cos(\nu\|\mathbf{k}\|_2)| / (2\nu\|\mathbf{k}\|_2 \sin(\nu\|\mathbf{k}\|_2))$ subject to exact asymptotic sound

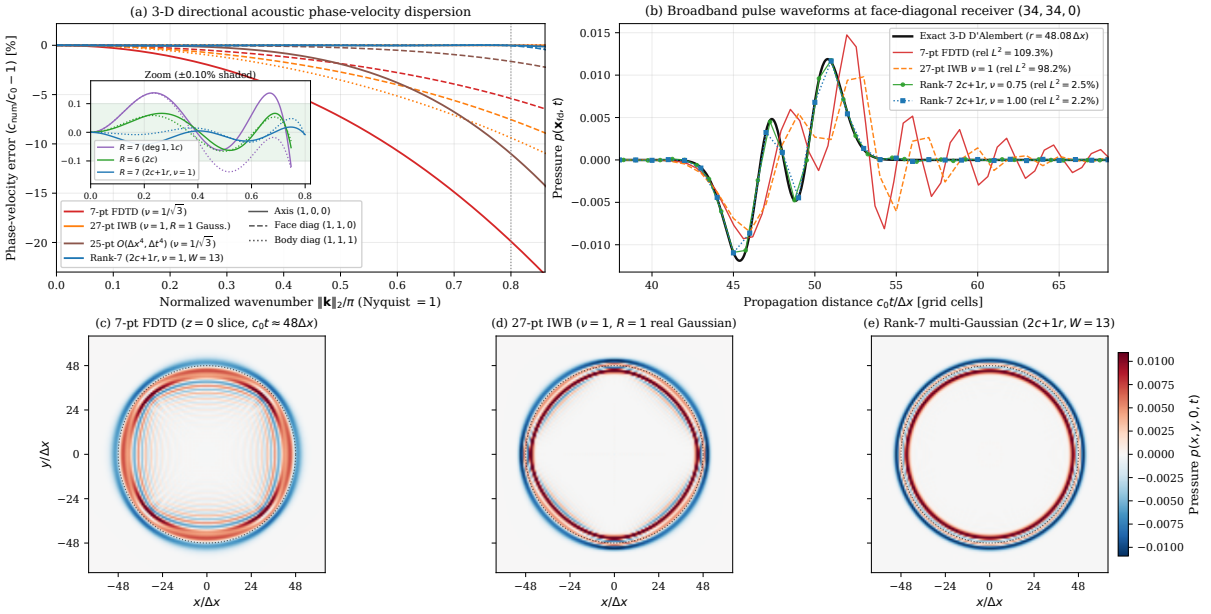


Figure 4: Three-dimensional acoustic wave propagation (`Applications/acoustics_3d.py`): (a) directional phase-velocity error ($c_{\text{num}}(\mathbf{k})/c_0 - 1$) along $(1,0,0)$, $(1,1,0)$, and $(1,1,1)$ across $\|\mathbf{k}\|_2/\pi \in [0, 0.86]$, with inset zoom ($\pm 0.10\%$ shaded) for $R = 6, 7$; (b) broadband pulse waveforms ($\sigma_0 = 1.40\,\Delta x$) on a 256^3 grid at the face-diagonal receiver $(34, 34, 0)$ ($r = 48.08\,\Delta x$) vs. the exact 3-D D'Alembert solution; (c–e) mid-plane wavefield slices $p(x, y, 0, t)$ at $c_0 t \approx 48\,\Delta x$ (dotted circle: $r = 48\,\Delta x$).

speed $\partial^2_{k_x} \widehat{M}(\mathbf{0}) = -2\nu^2$ and strict interior stability $-2 + 10^{-3} \le \widehat{M}(\mathbf{k}) \le 2 - \min(10^{-3}, \frac{1}{4}\nu^2 \|\mathbf{k}\|_2^2)$ across 232,620 symmetry-inequivalent Brillouin-octant and spherical-shell wavevectors (including the exact sphere $\|\mathbf{k}\|_2 = k_c$) is solved via cutting-plane HiGHS linear programming, followed by multi-start continuous Nelder–Mead maximization of $|\delta c(\mathbf{k})/c_0|$ over $0 < \|\mathbf{k}\|_2 \le k_c$. Moreover, because every 1-D radial factor $h[m] = h[-m]$ is even, half-sample symmetric index reflection at domain boundaries simultaneously diagonalizes all 3 directional passes in the 3-D `DCT-II` cosine basis—enforcing exact rigid-wall Neumann boundary conditions ($\partial_n p = 0$) without padding buffers or boundary instabilities ($\le 7.0 \times 10^{-7}$ max `float32` relative energy drift across all three `Rank-7` schemes over 1,000 steps in a 64^3 room; 2.0×10^{-7} at $\nu = 0.75$ and 6.1×10^{-8} at $\nu = 1.00$ for $2c + 1r$).

Table 10 and Figure 4 summarize accuracy and GPU timings across all five baselines (`DirectStep3D` is a tuned dense W^3 reflecting leapfrog Triton kernel; `SeparableStep3D` sweeps x-slabs sized to L2 cache in three passes per slab—x fan-out to R channels, y, and z fused with the leapfrog update—using the same 3×64 MiB field buffers plus 16–36 MiB of L2-resident slab scratch, 28 MiB for `Rank-7`):

- **Within separable spatial stencils:** Replacing naive $(n_{\text{re}} + 4n_{\text{cx}})$ expansion by Proposition 2.9's and 2.10's exact rank R cuts `float32` GPU step time by 24.0% on `Rank-7` (1.51 ± 0.03 ms vs. 1.98 ± 0.02 ms, $4.7\times$ faster than the tuned 13^3 direct Triton stencil; 15.6% on `Rank-4`, 23.5% on `Rank-6`, and 31.2% on `Rank-9`), alongside 21.2% on the 14-core CPU (283.6 ± 2.3 ms vs. 360.1 ± 0.6 ms on 128^3). On the 256^3 broadband spherical pulse test ($\sigma_0 = 1.40\,\Delta x$), all compact explicit stencils (7-pt, 13-pt, 27-pt IWB, and 25-pt $O(\Delta x^4, \Delta t^4)$) suffer 78.2%–109.3% waveform error from high-wavenumber dispersion, whereas `Rank-7` ($2c+1r$) keeps $|\delta c/c_0|$ at 0.045% ($\nu = 0.75$) and 0.049% ($\nu = 1.00$, < 1 cent) up to 0.80π (2.47 points per wavelength at $k_{0.1\%} = 0.81\pi$), achieving 1.6%–2.8% broadband waveform error across all axes and diagonals (2.8% with `float16` slab intermediates at $\nu = 0.75$). Consistent with the small-stencil regime in Table 7, freeing the $(L+1)$-tap even 1-D discrete factors

Table 10: Illustrative 3-D acoustic propagators vs. compact explicit FDTD (7-pt, 13-pt $O(\Delta x^4, \Delta t^2)$, 27-pt IWB, and fourth-order 25-pt $O(\Delta x^4, \Delta t^4)$ [31]) and L2-chunked periodic k-space `cuFFT` baselines on 256^3 voxels (explicit parameters in Table 12). Here $|\delta c/c_0|$ is evaluated on each row's target passband (0.80π for FDTD baselines, 1.00π for k-space `cuFFT`, and the parenthesized k_c for each propagator), whereas $k_{0.1\%}$, $k_{0.5\%}$, and Wave L^2 use identical definitions across all rows. GPU step times depend only on (W, R) (shared across the three $W = 13, R = 7$ rows) and report means over 10 CUDA-Event batches (6 batches for direct W^3 Triton; std ≤ 0.026 ms for separable, ≤ 0.045 ms for direct, 0.048 ms for k-space `cuFFT`, ≤ 0.0003 ms for FDTD; run-to-run variation $\leq 3\%$).

Scheme	ν	W	R (nv)	$\lvert\delta c/c_0\rvert$	$k_{0.1\%}$	$k_{0.5\%}$	Wave L^2	Direct	Naive	Step `fp32`	Step `fp16`
7-pt FDTD	0.58	3	—	19.88%	0.060π	0.135π	109.3%	—	—	0.324 ms	—
13-pt $O(\Delta x^4, \Delta t^2)$	0.50	5	—	8.74%	0.099π	0.222π	100.8%	—	—	0.324 ms	—
27-pt IWB ($1r$)	1.00	3	1 (1)	9.38%	0.085π	0.191π	98.2%	—	—	0.324 ms	—
25-pt $O(\Delta x^4, \Delta t^4)$	0.58	5	—	11.00%	0.236π	0.355π	78.2%	—	—	0.325 ms	—
k-space `cuFFT` (per.)	1.00	256	—	0.000%	1.00π	1.00π	0.35%	—	—	1.44 ms	—
Rank-4 ($1c{+}1r, 0.75\pi$)	0.75	9	4 (5)	0.293%	0.14π	0.76π	11.6%	2.68 ms	1.02 ms	0.857 ms	0.627 ms
Rank-6 ($2c, 0.75\pi$)	0.75	11	6 (8)	0.066%	0.76π	0.80π	5.2%	4.63 ms	1.66 ms	1.27 ms	0.920 ms
Rank-7 (deg 1 $1c, 0.75\pi$)	0.75	13	7 (9)	0.138%	0.16π	0.78π	7.1%	7.07 ms	1.98 ms	1.51 ms	1.09 ms
Rank-7 ($2c{+}1r, 0.80\pi$)	0.75	13	7 (9)	0.045%	0.81π	0.85π	2.8%	7.07 ms	1.98 ms	1.51 ms	1.09 ms
Rank-7 ($2c{+}1r, 0.80\pi$)	1.00	13	7 (9)	0.049%	0.81π	0.87π	2.5%	7.07 ms	1.98 ms	1.51 ms	1.09 ms
Rank-9 ($3c, 0.85\pi$)	0.75	17	9 (12)	0.044%	0.86π	0.89π	2.0%	14.5 ms	3.21 ms	2.21 ms	1.57 ms

from the continuous Gaussian formula under the same octant LP/Nelder–Mead objective substantially improves the smallest stencil (`Rank-4`, $W = 9, L + 1 = 5$: $0.293\% \to 0.108\%$) while yielding diminishing gains as L grows (`Rank-6`, $W = 11, L+1 = 6$: $0.066\% \to 0.056\%$).

- **Comparison with 1-pass compact FDTD (including fourth-order 25-pt) and global `cuFFT`:** On 256^3 voxels, all four fused 1-pass explicit FDTD kernels (7-pt, 13-pt, 27-pt IWB, and 25-pt $O(\Delta x^4, \Delta t^4)$) run at the DRAM-bandwidth limit of 0.324–0.325 ms/step (51.7–51.8 GVox/s, 92.5% of peak GDDR6X bandwidth)—now just 3.4× to 4.7× faster per step than L2-slab `Rank-7` (1.09 ms `fp16` / 1.51 ms `fp32`). When coarse dispersion is acceptable, compact 27-pt/25-pt schemes [42, 31] remain faster and simpler. Under an analytical $(\Delta x)^{-3}(\Delta t)^{-1} = \nu^{-1}(\Delta x)^{-4}$ resolution-scaling extrapolation, `Rank-7` at $\nu = 1.00$ ($k_{0.1\%} = 0.81\pi, k_{0.5\%} = 0.87\pi$) reduces voxel count by 865× and estimated runtime by about 1,800×–2,400× relative to 27-pt IWB ($\nu = 1.00, k_{0.1\%} = 0.085\pi$) at 0.1% tolerance (94× fewer voxels and about 90×–130× at 0.5%), while against fourth-order 25-pt $O(\Delta x^4, \Delta t^4)$ ($\nu = 1/\sqrt{3}, k_{0.1\%} = 0.236\pi, k_{0.5\%} = 0.355\pi$, including the $\sqrt{3}$ Courant-step factor) it uses 40× fewer voxels and an estimated 52×–71× less runtime at 0.1% tolerance (15× fewer voxels and about 13×–19× at 0.5%; at $\nu = 0.75$, about 39×–54× at 0.1% and 9×–13× at 0.5%). On a *homogeneous periodic or rectangular box* where global spectral transforms apply, our L2-chunked 256^3 periodic k-space `cuFFT` step [67] (`KSpaceStep3D`: 2-D real FFTs on 32-plane chunks followed by x-FFTs and spectral multiply on 16-row chunks) takes 1.44 ± 0.05 ms on GPU (83.3 ± 4.9 ms on the 14-core CPU) with zero phase-velocity dispersion up to π (0.35% pulse L^2 error)—remaining more accurate than any finite stencil, while running at essentially the same speed as `Rank-7` in `float32` (1.51 ms, within run-to-run noise), slower than `Rank-7` with `float16` intermediates (1.09 ms) and `Rank-4`/`Rank-6` (0.86/1.27 ms), and faster than `Rank-9` (2.21 ms). Local separable stencils remain primarily motivated where global all-to-all FFTs are undesirable (e.g. multi-GPU domain decomposition with L-cell halos), subject to the boundary and heterogeneity challenges below.

7 Limitations, Uncertainties, and Open Problems

To avoid overstating either the theoretical scope or the practical applicability of our results, we summarize six limitations, trade-offs, and remaining uncertainties:

1. **Continuous-domain exactness, discrete aliasing, and non-certified heuristic optimization:** Theorems 2.3–2.5 classify *exact* representations of *continuous* radial kernels on $\mathbb{R}^d$ (leaving $\varphi(s)$ for $s < 0$ unconstrained). Smooth non-exponential-polynomial targets ($e^{-\|x\|^4}$, Airy patterns $(J_1(r)/r)^2$, ideal circular low-pass filters) have infinite exact rank, while discontinuous kernels require separate arguments ($\mathbf{1}_{\{\|x\|=0\}}$ has rank 1; $\mathbf{1}_{r\le 1}$ has infinite rank). Sampling a continuous rank-R kernel on $\mathbb{Z}^d$ yields discrete rank *at most* R (strictly lower at aliased frequencies $\operatorname{Im}\lambda_u \in \pi\mathbb{Z}$ or when $W < R$). Moreover, while our inner Chebyshev weight solver is an exact linear program, outer optimization over nonlinear rates (α_u, ω_u) is nonconvex and heuristic: for $6c$ disc blur ($R = 12$), `blur_2d.py` reaches ripple ± 0.00193 whereas dedicated Remez exchange in [60] reaches ± 0.0014, and in 3-D acoustics our stability check ($|\widehat{M}(\mathbf{k})| < 2$ across 232,620 Brillouin-octant/shell points plus multi-start continuous Nelder–Mead searches) is a dense numerical audit rather than a formal interval-arithmetic or sum-of-squares proof over all $\mathbf{k} \in [-\pi, \pi]^3 \setminus \{\mathbf{0}\}$.

2. **Visual and rendering limitations in 2-D lens blur:** Truncating 1-D passes to $[-S, S]$ produces a square $[-S, S]^2$ support ($r \le \sqrt{2}\,S$), and flat-top approximations built from oscillatory complex Gaussians inevitably exhibit small negative stopband ripples ($\pm\delta$). Around extreme HDR impulses ($10^3\times$–$10^5\times$ scene white), negative lobes can create dark ringing halos unless clamped or tone-mapped, whereas positive Monte Carlo disc gathering is strictly non-negative. Separable convolution is also inherently shift-invariant: real depth-of-field rendering with spatially varying circle-of-confusion $R_{\text{px}}(x, y)$ suffers foreground/background occlusion bleeding unless paired with depth-layer decomposition [27], and for purely shift-invariant blurs at $R \ge 8$ or large radii ($R_{\text{px}} \ge 32$), complex-packed 2-D `cuFFT` (0.45–0.56 ms on 1920×1080 RGBA, Table 9; layout conversion between interleaved RGBA and two complex planes not timed, and tensor-core `float16` matrix-multiply formulations not explored) is faster than direct FIR cascades.

3. **Conditioning of root-of-unity identities at large d or high multiplicity k:** In Propositions 2.9 and 2.10, summing M alternating products with prefactor $\frac{2^{d-1}}{M}$ bounds worst-case term amplification for unit-modulus factors by 2^{d-1}: benign for $d = 2, 3, 4$ ($2^{d-1} = 2, 4, 8$), but exponential in $d \gg 1$. For degree $k = \deg P \ge 1$ on $[-S, S]^d$, evaluating $U_j(X)$ on $|X| = 1$ mixes $O(1)$ and $O(S^{2k}/k!)$ coefficients; rescaling $X \mapsto \rho X$ with $\rho \approx S^{-2}$ ($\rho^{-k}[X^k]\prod_j U_j(\rho X)$) balances monomial magnitudes, while balanced roots ($|\beta|^{1/d}$) equalize mode norms. Near coalescing rates (compare the $O(h^{-1})$ weights in Section 3.4), $d = 2$ permits exact *a posteriori* $\mathrm{GL}_R(\mathbb{R})$ orthogonalization across components on $\mathrm{rowSpan}(K)$ (Figure 2c), whereas the lack of transitive $\mathrm{GL}_R(\mathbb{R})$ cross-component rotations in $d \ge 3$ favors well-separated complex Gaussians ($k = 0$) over high-degree repeated rates.

4. **Wide-stencil obstacles, boundaries, hardware sensitivity, and global FFTs:** While `DCT-II` half-sample reflection handles homogeneous rectangular rigid rooms cleanly, wide stencils ($L = 4$–8 grid cells in `Rank-4`–`Rank-9`) reach across thin walls or baffles and cannot be applied directly at staircased, curved, or frequency-dependent impedance boundaries without one-sided boundary closures whose passivity and stability for signed multi-Gaussian stencils are currently unknown. On simple homogeneous boxes without interior obstacles, L2-chunked periodic `cuFFT` k-space propagation (1.44 ms/step, Table 10) is dispersion-free and matches `Rank-7 fp32` (1.51 ms) in speed (though slower than `fp16`-intermediate `Rank-7` at 1.09 ms and `Rank-4`/`Rank-6`). Moreover, all GPU timings reflect one Ada Lovelace GPU (48 MB L2 cache, 672 GB/s GDDR6X) under Triton `3.9.0` /

CUDA `13.2` with up to 3%–6% run-to-run autotuning variation: because `gpu_kernels.py` sizes 2-D strips and 3-D slabs explicitly to the 48 MB L2 cache, exact crossovers depend on L2 capacity, register file size, and kernel fusion (as illustrated by the 14-core CPU comparisons in Sections 6.3–6.4). Within the 3-pass-per-slab GPU kernel, replacing naive $n_{\mathrm{re}} + 4n_{\mathrm{cx}}$ by exact $n_{\mathrm{re}} + 3n_{\mathrm{cx}}$ reduces per-slab arithmetic and L2 traffic (24.0% step-time reduction on `Rank-7`), whereas on the unfused CPU baseline it reduces pass count directly (21.2% time reduction).

5. **Scope of formal verification and cross-literature connections:** Lean 4 and Isabelle/HOL guarantee that the $22 + 1$ formal theorems follow logically from their definitions and standard classical axioms, but human inspection remains essential to verify that those definitions (`IsSumOfProducts`, `separableRank`, `IsRepresentation`) faithfully capture the intended mathematical objects, and none of the steerable pass-count arguments, Python/Triton GPU code, or numerical tables in Section 6 are formally verified. Likewise, line-count comparisons reflect library state and author fluency rather than intrinsic proof length, and while (9) links our exact rank formulas directly to Kronecker ($d = 2$), Winograd–Alder–Strassen ($d = 3$), and Bläser–Mayer–Shringi ($d \geq 4$) algebra-complexity lower bounds [76, 1, 10], special cases of the explicit root-of-unity functional syntheses (Propositions 2.9–2.10) may also appear under different terminology in other communities.

6. **Open problem — sharpness of the general compact-open border-rank bound in $d \geq 3$:** Theorem 2.8 proves that every representation $\mathcal{R}$ has compact-open real border rank at most $B_d(\mathcal{R}) = C_d(\mathcal{R}) - (d-2)\sum_u \deg P_u$ (and complex border rank at most $\sum_{\mu\in S}(\deg P_\mu + 1) = \dim_{\mathbb{C}} \mathcal{A}_\psi$). For a single real degree-1 component $(0, X)$ ($K(x) = \|x\|^2$), $B_d = 2$ is sharp (W-state border rank 2 [8, 14]). Whether $B_d(\mathcal{R})$ equals the exact compact-open real border rank for all nonreal and higher-degree radial profiles in $d \geq 3$—for instance, whether $\|x\|^2\cos(\|x\|^2)$ can be approximated uniformly on compact sets below rank $2d$ (6 in $d = 3$) by non-Gaussian separable sequences (compare border-rank and border-subrank results on local algebras [80, 16])—remains open.

A Explicit Filter and Propagator Coefficients

To allow standalone reproduction of the main designs in Section 6 from this PDF (complete JSON archives for all 18 blur variants and all 6 acoustic propagators are provided in `Applications/`), every radial profile in Tables 11–12 is tabulated in the single canonical convention of Definition 2.2:

$$\begin{aligned}\varphi(s) &= \sum_{u=1}^{m} \mathrm{Re}\left(\left(\sum_{q=0}^{k_u}(p_{u,q} + iq_{u,q})s^q\right)e^{(-\alpha_u+i\omega_u)s}\right)\\ &= \sum_{u=1}^{m}\sum_{q=0}^{k_u} s^q\big(p_{u,q}\cos(\omega_u s) - q_{u,q}\sin(\omega_u s)\big)e^{-\alpha_u s},\end{aligned} \tag{16}$$

where real-rate components satisfy $\omega_u = 0$ and $q_{u,q} = 0$. In 2-D circular disc blur (Table 11), $s = (x^2 + y^2)/R_{\mathrm{px}}^2$ on $|x|, |y| \leq (1 + \Delta r)R_{\mathrm{px}}$ ($\Delta r = 0.2$), normalized in 2-D discrete execution so $\sum_{x,y} K[x, y] = 1$. In 3-D acoustic leapfrog propagation (Table 12), $M[\mathbf{m}] = \alpha_0\delta_{\mathbf{0}}[\mathbf{m}] + \varphi(m_x^2 + m_y^2 + m_z^2)$ on integer grid offsets $\mathbf{m} \in [-L, L]^3$ ($W = 2L+1$), where $\alpha_0 := 2 - \sum_{\mathbf{m}\in[-L,L]^3}\varphi(\|\mathbf{m}\|^2)$ must be computed from the evaluated φ grid so $\widehat{M}(\mathbf{0}) = 2$ holds to machine precision (avoiding exponential constant-mode blow-up $\widehat{M}(\mathbf{0}) > 2$ and preserving spatially constant initial states; note $\zeta = 1$ is a double root of $\partial_{tt}\bar{p} = 0$, so nonzero initial velocity mean gives physical $a + bn$ motion). Table 12 lists the exact α_0 recomputed from the 9-decimal coefficients (within

Table 11: Explicit canonical coefficients (16) for representative 2-D circular disc-blur designs ($\Delta r = 0.2$, Table 6).

Design ($R = C_2$, C_3, $\pm\delta$)	u (q)	α_u	ω_u	$p_{u,q}$	$q_{u,q}$
$1c$ (R=2, C_3=3, $\pm$0.2324)	1 (0)	0.862715	1.624792	+0.767553	−1.862917
$2c$ (R=4, C_3=6, $\pm$0.0759)	1 (0)	0.894640	5.265923	+0.413699	+0.555222
	2 (0)	1.973219	1.554245	+0.510373	−4.611414
$1c_2$ (R=4, C_3=7, $\pm$0.1165)	1 (0)	2.878934	3.152485	+0.883521	−7.976371
	1 (1)	2.878934	3.152485	−16.607444	+7.931853
$2c$+$1r$ (R=5, C_3=7, $\pm$0.0446)	1 (0)	0.962816	7.154625	+0.091824	−0.456790
	2 (0)	2.075279	3.285822	−3.434558	+0.588375
	3 (0)	2.494577	0	+4.387353	0
$4c$ (R=8, C_3=12, $\pm$0.0109)	1 (0)	4.338835	1.553577	−5.754013	−46.161778
	2 (0)	3.838091	4.692138	+9.776587	+15.218676
	3 (0)	2.790788	8.181910	−3.042741	−0.293993
	4 (0)	1.345593	12.330086	+0.009312	−0.245279
$6c$ (R=12, C_3=18, $\pm$0.0019)	1 (0)	5.143786	2.080541	−82.448179	−111.252655
	2 (0)	5.613157	6.153062	+113.961587	−58.149957
	3 (0)	5.982623	9.803250	+39.766911	+161.948149
	4 (0)	6.506533	11.060675	−71.545624	−94.857314
	5 (0)	3.871248	14.810010	+1.416029	+3.711177
	6 (0)	2.202364	19.033594	−0.152648	+0.108514

2.2×10^{-8} of `float64` JSON), matching $\partial_{k_x}^2 \widehat{M}(\mathbf{0}) = -2\nu^2$ to $\le 8 \times 10^{-8}$ and keeping $\widehat{M}(\mathbf{k}) \in [-1.9990, 1.9946] \subset (-2, 2)$ on the verification grid.

Code and Data Availability

All source code, machine-checked proofs, numerical optimization scripts, GPU benchmarks, JSON parameter/result archives, and figure-generation scripts for this work are openly available in the public domain (CC0 1.0 Universal) at `https://github.com/OlliNiemitalo/kernel-rank-lean`:

- `KernelRankLean/` and `Experiments/`: Lean 4 (`v4.36.0-rc2` with Mathlib `v4.36.0-rc2`) formalizations of all d-D and 2-D separable-rank, root-of-unity, differential-annihilator, border-rank, and alternative semigroup theorems;
- `Isabelle/`: Isabelle2026 (`HOL-Nonstandard_Analysis`) formalization `Semigroup_NSA.thy`;
- `Applications/`: Python, SciPy, PyTorch, and Triton optimization, GPU-kernel, and figure scripts (`blur_2d.py`, `acoustics_3d.py`, `gpu_kernels.py`, `generate_figures.py`) plus JSON result archives (`results_blur_2d.json`, `results_acoustics_3d.json`);
- `scripts/`: hermetic setup (`scripts/setup`), formal verification and axiom/oracle audit (`scripts/check`), isolated GPU Python wrapper (`scripts/python`), and PDF build (`scripts/build-paper`).

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This work builds directly on problems, mathematical formulations, and software developed by Olli Niemitalo, including `kernel-optimizer` [60] for complex-Gaussian approximations of 2-D circularly symmetric blur kernels and his investigations of isotropic low-pass filters, steerable gradient orientation estimation, and planar harmonic differential operators [56, 57, 58, 59]. Olli Niemitalo initiated and guided the `kernel-rank-lean` repository, proposed comparing standard and Isabelle/HOL nonstandard proofs of the semigroup step, suggested extending

Table 12: Explicit 9-decimal canonical coefficients (16) for all six 3-D acoustic leapfrog propagators in Table 10 (set $\alpha_0 := 2 - \sum_{\mathbf{m}\in[-L,L]^3} \varphi(\|\mathbf{m}\|^2)$; reference α_0 shown for verification).

Scheme (ν, W, R, α_0)	u (q)	α_u	ω_u	$p_{u,q}$	$q_{u,q}$
Rank-4 (1c+1r, ν=0.75, W=9, R=4, α_0= − 1.440427676)	1 (0)	0.439979390	0.502750129	+0.137876459	+0.024929224
	2 (0)	0.888101329	0	+0.492695925	0
Rank-6 (2c, ν=0.75, W=11, R=6, α_0= − 1.419560209)	1 (0)	0.606539114	0.240982835	+0.597581203	+0.180225818
	2 (0)	0.267859019	0.385016560	−0.007190817	+0.050272529
Rank-7 (deg 1 1c, ν=0.75, W=13, R=7, α_0= − 1.434649594)	1 (0)	0.429753880	0.229296784	+0.605530664	+0.525638174
	1 (1)	0.429753880	0.229296784	−0.061586407	−0.061209821
Rank-7 (2c+1r, ν=0.75, W=13, R=7, α_0= − 1.439243312)	1 (0)	0.381717837	0.311635272	+0.110431944	+0.263079509
	2 (0)	0.163094730	0.300649461	−0.015761751	−0.008519990
	3 (0)	0.599866965	0	+0.491386225	0
Rank-7 (2c+1r, ν=1.00, W=13, R=7, α_0= − 1.959018821)	1 (0)	0.321809087	0.262377242	+0.159170237	+0.129231489
	2 (0)	0.126172334	0.278598443	−0.010612383	+0.004174471
	3 (0)	0.438442015	0	+0.241253606	0
Rank-9 (3c, ν=0.75, W=17, R=9, α_0= − 1.447076570)	1 (0)	0.301382846	0.246772855	−0.286641369	+0.234406754
	2 (0)	0.398975160	0.094363260	+0.866865360	+0.865008109
	3 (0)	0.111967878	0.275076161	−0.002296910	−0.011770692

the formalization to d dimensions alongside the visual pipeline exposition and 2-D/3-D GPU applications, and shared independent related-work reviews that sharpened the commutative-algebra and operator-theory connections. We also thank the aiXiv Official Agent reviewers on versions 1.0, 1.1, 1.2, and 1.3, whose critiques motivated Theorems 2.5 and 2.8, Proposition 2.10, the intuitive cost guide and $k = 1$ worked example, the `separableChannels` ablation, the notation harmonization, Section 7, and Appendix A. Both the repository and this manuscript are released under CC0 1.0 Universal.

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